ISSUE 01 / NOVEMBER 2026

JOTNAR FIELD MANUAL

FIELD MANUAL



Streams, layouts, exact integers, and the machines that witness them.

INSIDE THIS ISSUE

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+	THE 86-MNEMONIC REGISTER

THE DETERMINISTIC PRESS / THE JOTNAR WORKING GROUP

JOTNAR-LAN · JOTNAR-MAC · JOTNAR-WIT

ISSUE 01 / DIGITAL / FIRST ROUGH DRAFT

NOVEMBER 2026

JOTNAR FIELD MANUAL

THE GEOMETRIC ASSEMBLY LANGUAGE

EDITOR
The Jotnar Working Group

ORACLE
The Python reference implementation

MACHINES
C99 · non-FPU · CUDA · Verilog · RV32IM

WITNESS
Differential parity, byte for byte

TYPE
Anton · Barlow Condensed · Courier Prime · IBM Plex Mono · Stardos Stencil

SETTING
Markdown, Paged.js, Chromium

FAIL-LOUD NOTICE. Unknown mnemonics, arity mismatches, use-before-def, layout conflicts, and shape mismatches are errors, never silent guesses. If this manual is wrong, the assembler is the authority.

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JOTNAR FIELD MANUAL

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ISSUE 01
DIGITAL

EVERYTHING IN THIS ISSUE

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The manual is a graph too. Follow the part labels, or jump straight to the register.

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THE DETERMINISTIC PRESS

00

ORIENTATION

What this is, who it is for, and the three ideas the rest of the book keeps using.

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00 READ THIS FIRST

This is a first rough draft of a manual for a language that is still being built. That is the point.

You are holding a field manual for **Jotnar**, a geometric assembly language for typed streams. It is not a tutorial for a finished product. It is a working document for a language that is being ported, tested, and argued with in public.

The code is real. The examples run. The edges are still sharp.

Jotnar is the family/stack. This manual documents the Rust implementation of **Jotnar-LAN/DIG**: a program format, a reference interpreter, and a set of emitters for C, integer-only C, CUDA, Verilog, and an RV32IM machine. The Python implementation is the oracle. This manual follows the same doctrine as the code: **unknown mnemonics, arity mismatches, use-before-def, layout conflicts, and shape mismatches are errors, never silent guesses.**

HOW TO READ THIS BOOK

There are three passes. **First pass:** read the manifesto, then the geometry chapter, then the four first listings. You will be able to read a Jotnar listing without running it. **Second pass:** read the instruction-set chapters with the register open. **Third pass:** read the machine chapters and build something. The appendices are for when you want to change the book itself.

THE FIELD KIT

You need a terminal, a Rust toolchain, and a payload. The payload is a JSON file that binds names to typed streams. That is the whole boundary. There is no hidden state, no environment variable that changes arithmetic, no configuration file that silently rescales your data.

```
# build the reference implementation
cargo build --release

# check, verify, run
./target/release/jotnar check  showcase/language/saxpy.asm
./target/release/jotnar verify showcase/language/saxpy.asm
./target/release/jotnar run    showcase/language/saxpy.asm \
  --payload /tmp/saxpy.json --outputs OUT
```

The payload for that run is small enough to read in one sitting:

```
{"streams":{
  "A":{"kind":"I64","shape":[4],"data":["3","3","3","3"]},
  "X":{"kind":"I64","shape":[4],"data":["1","2","3","4"]},
  "Y":{"kind":"I64","shape":[4],"data":["10","20","30","40"]}
}}
```

The output is `[13, 26, 39, 52]`. If you got that, the rest of the book is elaboration.

PREREQUISITES

NEED	FOR
Rust stable + Cargo	everything.
gcc	the C backend, the native DOOM core, and the Verilog fixture.
CUDA toolkit + GPU	the CUDA target and GPU migration parity.
SDL2 runtime	the playable DOOM window.
DOOM image + WAD	live DOOM; run_doom_demo.sh falls back otherwise.
Python 3 + NumPy/Pillow	the parity harnesses against the Python oracle.
jotnar-main + phi-core checkout	differential parity only; the demos run without them.

`run_doom_demo.sh` is **playable by default.** `--frames N` is the headless path. If the external image or WAD is absent, the script falls back to the checked-in reference title and does not pretend the live run happened.

DRAFT NOTICE

This is **issue 01**, a first rough draft. Some chapters describe behavior that is gated by parity scripts but not yet exposed in the CLI. Some commands are stable; some are scaffolding. Where the manual and the assembler disagree, the assembler wins. Where the assembler and the oracle disagree, the parity harness wins. Where the harness and reality disagree, file a bug.

STATUS AT THIS DRAFT / 2026-11-06

The Rust workspace is green at **61/61 tests**. The self-contained trust-spine example verifies **PASS**. The real DOOM certificate verifies **PASS on this host**, and correctly prints **BLOCKED** when its local `/tmp` artifacts are absent. The manual's witness script passes **4/4** listings. The trust spine has its own appendix now: **Appendix A**.

CONVENTIONS

CONVENTION	MEANING
UPPERCASE	A mnemonic, directive, or keyword.
lowercase	A stream name chosen by the author.
$VAR	A layout variable, unified by the assembler.
KIND:GEOM	A concrete layout, such as T:HW or F:HWC.
*	A wildcard layout: matches anything, binds nothing.
→	“produces” in a signature; 2 → 1 means two inputs, one output.
bold	A term defined in the glossary.
#	A comment, to the end of the line.

Jotnar listings are line-oriented. An assignment names one or more outputs on the left and one operation on the right. There are no registers, no stack, and no instruction pointer. The listing order is for the reader; the **dataflow** is the program.

THE FAIL-LOUD PROMISE

Most languages try to be helpful when you make a mistake. Jotnar tries to be honest. A wrong arity is an error. A layout conflict is an error. A use-before-def is an error. A shape mismatch at run time is an error. An integer division by zero is an error. A float that is NaN or infinite where an integer is required is an error.

This is not masochism. It is the only way to keep a language that lowers to five different machines honest. If the frontend guesses, the emitters inherit the guess, and the witness is worthless.

IF THE ASSEMBLER HAS TO GUESS, THE PROGRAM IS ALREADY WRONG.

THE HOUSE RULE

WHAT IS NOT HERE YET

The manual describes the parts that exist. The honest gaps are also part of the documentation:

- the `jotnar-main` **device** emitter and the CUDA fleet kernel used by the original DOOM GPU demo;

- TCP transport and serve mode (`live=` resident loops), packed outputs, the codec ABI, CUDA graphs, tracing summaries, causal `overrides`, and the store-hook fused executor;
- the model-side phi-core modules (`calibrate`, `score`, `compose`, `stamp`, `zoo`);
- tuned CUDA kernels and cuBLAS scheduling. The current CUDA backend is correctness-first: one launch per operation;
- the Rust SDL player's intermittent frame-time spikes. It is playable; it is not yet smooth. A frame-time trace and a p99/max-frame gate are follow-ups, not core-exactness gaps.

The trust spine is **not** on this list. It is here: Appendix A. If you need one of the remaining gaps, you are reading the right draft at the wrong time. The next issue is the one to watch.



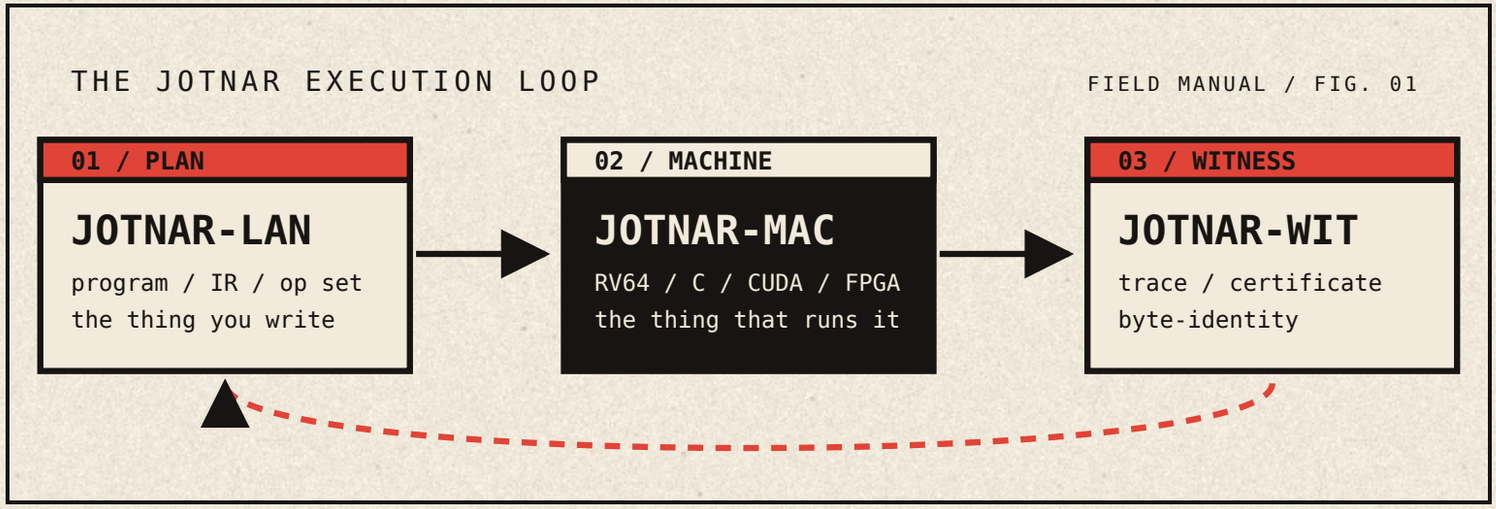
END

01 THE JOTNAR MANIFESTO

Jotnar is not one object. It is an execution loop: a program, a machine, and a witness.

Jotnar is a family of ideas about computation. The first idea is that a program can be a **graph of named streams** rather than a sequence of instructions that mutates a machine. The second is that the graph can be lowered to very different machines without changing its meaning. The third is that the result can be **witnessed** — compared, byte for byte, against an independent oracle.

Those three ideas have names. Use them.



The execution loop. The plan is not the machine, and the machine is not the proof.

THE THREE ASPECTS

ASPECT	WHAT IT IS	WHAT IT IS NOT
Jotnar-LAN	The language, IR, op set, and plan formats. The program.	A particular file extension or a Python module.
Jotnar-MAC	The machine or target: ISA, memory, lanes, OS, board.	The program itself.
Jotnar-WIT	The witness: trace, certificate, provenance, byte-identity.	A proof in the mathematical sense.

When the distinction matters, write the suffix: **Jotnar-LAN/DIG**, **Jotnar-MAC/DIG**, **Jotnar-WIT/DIG**. When it does not, "Jotnar" means the family. Do not use "Jotnar" as the name of a project. Do not call a neural network "Jotnar." The network is host code. Jotnar is the thing the host hands a plan to.

THE SHORT VERSION

Lachesis measures the deterministic thread; **Atropos** makes the accurate cut. **Jotnar-LAN** → **Jotnar-MAC** → **Jotnar-WIT** is the execution loop. **Field** → **Map** → **Vector** → **Plan** → **Certificate** is the data/compute loop. **DIG / SIM / ANA / HYB** says what kind of Jotnar you mean.

FIELDS ARE STATE. MAPS ARE STORED.

Storage and processing are different axes. Confusing them produces bad sentences and worse programs.

TERM	AXIS	MEANING
Jotnar-FLD (Field)	State	The spatial values/lattice: depth, normals, IDs, provenance, phi-core state. The *what*.
Jotnar-MAP (Map)	Storage	A stored field/artifact: shape, dtype, layout, hash. Data at rest.
Refined Jotnar Map (RJM)	Storage + provenance	A map that passed a refinement stage, carrying an error bound and provenance. Still data.
Jotnar-VEC (Vector)	Processing	A costed, data-parallel batch of work. Compute in motion.
Jotnar-LANE	Resource	The execution resource a vector runs on.
Jotnar-PLN (Plan)	Binding	The executable artifact that binds maps (inputs/outputs) to vectors (work).
Jotnar-CERT	Witness	The output of executing a plan.

FIELDS ARE STATE. MAPS ARE STORED. VECTORS ARE EXECUTED. PLANS BIND THEM. CERTIFICATES WITNESS THE RESULT.

THE DATA RULE

FOUR FLAVORS

Bare "Jotnar code" is ambiguous. Require a flavor.

FLAVOR	MEANING	STATUS

Jotnar-DIG	Digital: exact, bit-identical, C/CUDA/RV64, JMV1/JSW1.	Exists.
Jotnar-SIM	Simulated analog: DAC/RC/ADC, 2D field, digital twin.	Exists as experiments.
Jotnar-ANA	Physical analog: RC/crossbar/optical substrate, calibration and error bounds, no bit-exactness.	Does not exist.
Jotnar-HYB	Hybrid: digital control + analog kernel + digital readout/gate.	Not built; realistic target.

The RC field in the analog work is **Jotnar-SIM**, not **Jotnar-ANA**. The board is **Jotnar-MAC/DIG** running a **Jotnar-LAN/DIG** plan and emitting a **Jotnar-WIT/DIG** certificate. Say that, or say less.

WHY GEOMETRY?

Because geometry is the part of the problem that does not change when the machine does. A depth map is still a depth map on a CPU, a GPU, a piece of FPGA fabric, or a simulated analog field. A layout like `T:HW` is a promise about shape and kind that survives lowering. An exact integer add is exact on every target. A stream is a name for a value that does not care where it lives.

The language is small on purpose. There are 86 mnemonics. The registry is the authority. Every mnemonic has an arity, a layout signature, and a reference implementation. If a target cannot lower a mnemonic, it says so. If a shape rule fails, the runtime says so. If an emitter has no pattern, it returns `NoPattern`, loudly.

THE WITNESS RULE

The board is a witness, not a proof. Byte-identity against an oracle is strong evidence that the lowering preserved the plan. It is not a theorem that the plan computes what you meant. That part is still your job.

WHAT JOTNAR IS NOT

Jotnar is not a project name. It is not the depth model, the JBU, the normal net, or the layer net; those are **Atropos** host code. It is not a file format, a Python package, or a particular GPU kernel. It is the stack: a program, a machine, and a witness. The names are load-bearing. Use them carefully and the rest of the system stays reviewable.

END

THE DETERMINISTIC PRESS

THE LANGUAGE

A program is a graph of named streams. Learn to read the graph before you learn the instructions.

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02 THE GEOMETRY OF A PROGRAM

A Jotnar listing is a graph wearing the costume of a list. Learn to see the edges.

Start with the smallest useful program in the language:

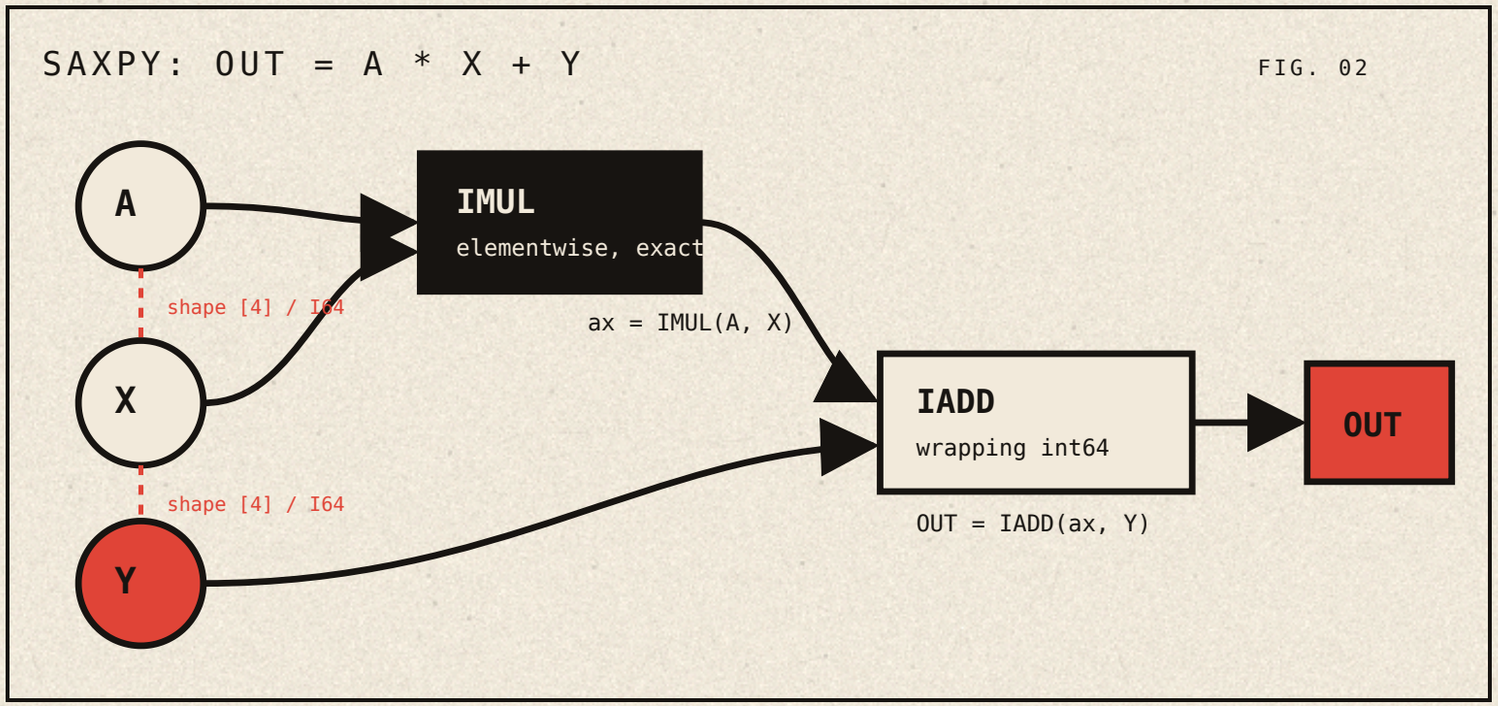
SAXPY / SHOWCASE/LANGUAGE/SAXPY.ASM

```
# SAXPY: A*X + Y.  A linear elementwise specimen.
IN A
IN X
IN Y

ax = IMUL(A, X)
OUT = IADD(ax, Y)
```

That is a complete listing. It declares three input streams, multiplies `A` by `X` elementwise, adds `Y`, and names the result `OUT`. The `#` lines are comments. The blank lines are for humans. The indentation is optional.

The important thing is not the order of the lines. It is the **shape of the graph**:



SAXPY as a dataflow graph. The names are the edges; the operations are the nodes.

`A`, `X`, and `Y` are sources. `IMUL` consumes `A` and `X` and produces `ax`. `IADD` consumes `ax` and `Y` and produces `OUT`. There is no register named `ax`, no stack slot, and no instruction pointer. `ax` is a name for a stream. If you never use it again, it is still a stream; the interpreter keeps it because another instruction might.

THE THREE VERBS

The CLI has three verbs for a listing, and they answer different questions.

```
# Does it parse, expand, bind, and check arity?
jotnar check showcase/language/saxpy.asm

# Can the declared layouts be unified? What is the DEF interface?
jotnar verify showcase/language/saxpy.asm

# Run it against a payload.
jotnar run showcase/language/saxpy.asm \
  --payload /tmp/saxpy.json --outputs OUT
```

`check` is syntax and registry binding. `verify` replays layout unification over declared layouts only. `run` executes the graph. They are deliberately separate: a program can bind and run without declaring a single layout, and a program can declare layouts that `verify` resolves even if the runtime shape checks would later reject the data.

> **CHECK, VERIFY, RUN**
>
> `check` says the listing is well-formed. `verify` says the declared layouts are consistent. `run` says the payload has the right shapes and the operations accept them. None of the three is a proof that the program computes what you meant.

For SAXPY, `check` reports `2 instructions, 3 inputs, 0 state stream(s)`. `verify` reports `layouts resolved: 2/5 (0 error(s))` because only `ax` and `OUT` have layouts that can be resolved without declared input layouts. That is honest coverage reporting, not a failure.

WHAT THE INTERPRETER DOES

`run` reads the payload, binds each declared input name to a typed stream, and executes the instructions in listing order. Because every instruction consumes only streams that already exist, listing order is a valid topological order. The interpreter does not need a scheduler. It does not need to know that `ax` comes before `OUT`; the assembler already rejected the listing if it did not.

The payload is ordinary JSON:

```
{"streams":{
  "A":{"kind":"I64","shape":[4],"data":["3","3","3","3"]},
  "X":{"kind":"I64","shape":[4],"data":["1","2","3","4"]},
  "Y":{"kind":"I64","shape":[4],"data":["10","20","30","40"]}
}}
```

The output is a stream map in the same shape. With `--outputs OUT`, the CLI writes only the named stream:

```
{"streams":{"OUT":{"kind":"I64","shape":[4],
  "data":["13","26","39","52"]}}}
```

Integer data is printed as strings so that values above 2^53 survive JSON. If you are reading this in a language with a native int64, you may parse them as numbers.

THERE ARE NO REGISTERS

This is the mental model that matters. A register is a place. A stream is a name for a value. The language has no place. It has names, operations, and shapes.

That has consequences:

- **Order is a convenience.** The listing is written in a valid order, but the meaning is the graph.
- **Reuse is a name choice.** If you write `x = IADD(x, 1)`, you are rebinding the name, not mutating a register. The old value is still available to any instruction that already consumed it. *(Some targets, notably Verilog, reject reassignment because hardware wires are not variables. More on that later.)*
- **Dead names are allowed.** A stream that is never consumed is still computed by the reference interpreter. The emitters may eliminate it; the language does not require them to.
- **Shapes travel with values.** A stream is not just a kind; it has a shape. A scalar is a stream with an empty shape. A vector is one dimension. An image is `H×W×C`.

THE LISTING IS A CONVENIENT SERIALIZATION OF A GRAPH. IF YOU CAN DRAW THE GRAPH, YOU CAN READ THE PROGRAM.

THE GRAPH RULE

THE FAIL-LOUD MOMENT

Try changing the arity:

```
IN A
IN X
OUT = IMUL(A)
```

The assembler refuses it. `IMUL` wants two inputs; you gave one. It does not insert a zero, broadcast a scalar, or hope for the best. The error names the operation and the source line. That is the contract.

The same contract applies to shapes. `IADD` requires its operands to have the same shape (or one to be a scalar literal). If you feed a `[4]` stream and a `[3]` stream, the runtime raises a shape mismatch. It does not truncate to the shorter one.

END

03 STREAMS, KINDS, AND SHAPES

A stream is a name, a kind, a shape, and a layout. The first three are data; the fourth is a promise.

Every value in Jotnar is a **stream**. A stream has four pieces of identity:

1. a **name** chosen by the author;
2. a **kind** — what the elements are;
3. a **shape** — how many elements and how they are arranged;
4. a **layout** — a compact string that combines kind and geometry for the assembler.

The name is local to the listing. The kind and shape are checked at run time. The layout is checked at assembly time, when it is declared.

THE KINDS

KIND	LAYOUT LETTER	PAYLOAD KIND	WHAT IT HOLDS
F32	F	F32	32-bit floats.
F64	F	F64 or F	64-bit floats.
I64	I	I64 or I	Signed 64-bit integers.
U8	U8	U8	Unsigned bytes.
Bool	I	Bool	Boolean masks.
Triple	T	T	Three planes: sign, exponent, zero flag.
None	UNKNOWN	None	Absence, used for history streams.

`F32` and `F64` are distinct even though both use the layout letter `F`. The reference interpreter preserves the distinction where numpy does. A `Bool` stream is an integer mask at the layout level; `SELECT` and `ISELECT` accept it.

`Triple` is the important odd one. It is not a float, and it is not an integer. It is a scale-free value in the phi lattice. Chapter 08 is entirely about it.

SCALARS

A scalar is not a separate kind. It is a stream with an empty shape. A bare number in argument position is a scalar literal. The layout resolver treats it as `F:SCALAR`. Operations that accept a scalar (shifts, clip bounds, axes, strides) read it as an integer or float depending on the operation.

SHAPES

SHAPE	MEANING	EXAMPLE LAYOUT
`[]`	Scalar.	`F:SCALAR`
`[N]`	Vector of length N.	`F:SEQ, I:*`
`[H, W]`	2D field or matrix.	`T:HW, F:HW`
`[H, W, C]`	Image with channels.	`F:HWC, U8:HWC`
`[B, H, W, C]`	Batch of images.	`*`
`[R, D]`	Rows of features.	`*`

Jotnar does not have a fixed list of shape names. `HW`, `HWC`, `SEQ`, and `SCALAR` are conventions used by the registry's signatures. `*` matches any shape and binds nothing.

SHAPE IS NOT OPTIONAL

The interpreter checks shapes for the operations that have shape rules: elementwise arithmetic, `MATMUL`, `WARP`, `SELECT`, `ARGMAX`, `GATHER`, and `GATHER3`. An operation without a shape rule can still fail loudly on its own terms. Never assume a wildcard in a signature means "any shape will work."

DECLARING INPUTS

An input declaration may carry a layout:

```
IN rgb AS U8:HWC
IN depth AS T:HW
IN pos AS I:SEQ
```

The layout is a promise that the assembler uses for unification. It is not a conversion. If the payload delivers an `F64` stream under a name declared `U8:HWC`, the runtime cross-check fails. The layout tells the assembler what to expect; the payload has to agree.

You can declare several names at once:

```
IN a, b AS T:SEQ
```

Each name gets the same layout. Duplicate input names are an error.

THE PAYLOAD

The payload is a JSON object with a `streams` map. Each stream has a `kind`, a `shape`, and data. The data key depends on the kind:

```
{
  "streams": {
    "x":   {"kind":"F64","shape":[2,3],
            "data":[0.1,0.2,0.3,0.4,0.5,0.6]},
    "pos": {"kind":"I64","shape":[2],"data":["0","1"]},
    "rgb": {"kind":"U8","shape":[2,2,3],"data":[0,1,2,3,4,5,6,7,8,9,10,11]},
    "t":   {"kind":"T","shape":[2,2],
            "s":[1,-1,1,1],"e":[30000,31000,32000,33000],"z":[0,0,0,0]}
  }
}
```

`F32` and `Bool` are accepted too. Integer entries may be numbers or strings; strings preserve int64 values above 2^53. The output uses the same shape, with integer values encoded as strings for the same reason.

> **NONE**
>
> A stream may be `None`. The `WARP` operation passes `None` through, which is how a temporal pipeline represents "no history yet." `None` is not zero. It is a distinct value that most operations reject.

KINDS ARE NOT COERCED

There is no implicit cast from `F` to `I`. `F2I` is an explicit operation, and it truncates toward zero, rejects NaN and infinity, and rejects values outside the int64 range. `I2F` is explicit too, and produces `F64`. If you want a different kind, ask for it.

The same is true of `Bool` and `U8`: they are distinct at the payload boundary even though many operations accept both as integer-like masks or bytes. The fail-loud doctrine is not only about errors. It is about making every conversion visible.

END

04 LAYOUTS: GEOMETRY AS TYPE

A layout is a small string that tells the assembler what a stream is shaped like — and lets it refuse the wrong graph before the runtime sees it.

A layout is a string of the form `KIND:GEOM`. `KIND` is one of `F`, `T`, `I`, `U8`, or `UNKNOWN`. `GEOM` is a shape convention such as `SCALAR`, `SEQ`, `HW`, or `HWC`. The full layout is a compact type for a stream.

```
F:SCALAR      a float scalar
F:SEQ         a float vector
F:HWC         a float image with channels
T:HW          a triple field
I:*           an integer stream of any shape
U8:HWC        a byte image
```

The registry gives every mnemonic a **layout signature**: one pattern per input, one per output. The assembler tries to unify the patterns against the actual layouts. If it cannot, the program is rejected. If it can, every output gets a resolved layout, or `UNKNOWN` when the information is not there.

THE PATTERN GRAMMAR

PATTERN	MEANING
`*`	Matches anything. Binds nothing.
`$VAR`	Unifies a whole layout string. The same variable must get the same layout everywhere.
`$VAR^T`	The transpose-derived form of `$VAR`.
`KIND:GEOM`	A literal layout. Must match exactly.

A simple elementwise operation uses one variable for all three positions:

```
ADD   $A, $A -> $A
```

If you call `ADD(T:HW, T:HW)`, the assembler binds `$A = T:HW` and gives the output `T:HW`. If you call `ADD(T:HW, T:HWC)`, it reports a **layout conflict**: `$A` was already `T:HW`, and the second argument wants `T:HWC`.

CONFLICT VS. MISMATCH

A **layout conflict** means two uses of the same variable disagree. A **layout mismatch** means a literal pattern does not match the actual layout. Both are assembly errors. Neither is repaired by broadcasting, truncation, or a silent cast.

TRANSPOSE

`TRANSPOSE` is the clean example of a derived layout:

```
TRANSPOSE   $A -> $A^T
```

If the input is `T:HW`, the output layout is `T:HW^T`. The actual operation swaps the last two dimensions, so the data is `W×H`; the `^T` suffix records that the layout was derived by transposition rather than declared independently. If the input is `UNKNOWN`, the output stays `UNKNOWN`.

WHY THE SUFFIX?

The suffix is not a separate shape language. It is a marker that the output was derived from the input by transposition. This lets the assembler keep the relationship between the two streams even when the concrete geometry is not known yet.

LITERAL SIGNATURES

Some operations are specific about kind and geometry:

SIGNATURE	MEANING
U8:HWC -> F:HWC	SRGB_DECODE: bytes in, floats out, same image shape.
F:HW -> T:HW	SQRT: float field in, triple field out.
F:HW2 -> I:HW	STATIC: a two-channel flow becomes a one-channel mask.
I:*, * -> I:*	IADD: an integer stream and a scalar-or-stream operand.

The literal patterns are how the registry prevents nonsense like decoding sRGB from a triple field or computing a flow mask from a scalar.

WILDCARDS ARE NOT PROMISES

`MATMUL` is declared as `*, * -> *`. It deliberately does not unify layouts, because matrix multiplication cares about the inner dimension, not the spelling of the geometry. The runtime has a separate shape rule:

```
a.shape[-1] == b.shape[-2]
```

If you forget the transpose, the runtime says:

```
inner dims [B,M,K] vs [B,N,K] (did you forget TRANSPOSE?)
```

That is the division of labor. The assembler handles the relationships it can see. The runtime handles the data-dependent ones. Both are allowed to fail loudly.

VERIFICATION COVERAGE

`jotnar verify` replays layout unification over declared layouts only. It prints a coverage line and, optionally, a DEF interface table.

```
$ jotnar verify showcase/language/saxpy.asm
layouts resolved: 2/5 (0 error(s))
```

Two of five streams have resolved layouts: `ax` and `OUT`. `A`, `X`, and `Y` have no declarations, so they are `UNKNOWN`. That is not an error; it is a coverage number. If you want a higher number, declare the inputs:

```
IN A AS I:SEQ
IN X AS I:SEQ
IN Y AS I:SEQ
```

Then `verify` can resolve all five.

UNKNOWN IS HONEST

`UNKNOWN` means the assembler does not have enough information. It is not a wildcard, and it is not a failure by itself. A program can run with unknown layouts; it just gets less static checking. The runtime shape checks are still active.

A LAYOUT IS A PROMISE THE ASSEMBLER CAN CHECK BEFORE THE MACHINE RUNS. A SHAPE IS A FACT THE RUNTIME CHECKS BEFORE THE OPERATION RUNS.

LAYOUT RULE

A WORKED DISAGREEMENT

Consider:

```
IN a AS T:SEQ
IN b AS T:HW
OUT = ADD(a, b)
```

The signature of `ADD` is `$A, $A -> $A`. The first argument binds `$A = T:SEQ`. The second wants `T:SEQ` but gets `T:HW`. The assembler reports a layout conflict at the `ADD` line. It does not add a sequence to a field, and it does not pretend the field is a sequence.

That is the whole type system in one example. It is small, but it catches a large class of mistakes before a payload is ever loaded.

END

05 COMPOSITION: DEF, CALL, IMPORT, REPEAT

A language of graphs still needs a way to name a subgraph, reuse it, and loop over it. Jotnar keeps all three explicit.

Composition in Jotnar is textual, not runtime. `DEF` defines a subgraph. `CALL` expands it at the call site, with fresh names for every internal stream. `IMPORT` splices another file. `REPEAT` duplicates a body or emits a runtime loop. None of these features is a function call in the C sense. They are ways to build a larger graph from smaller graphs.

DEF AND CALL

A definition names its formals, its outputs, and a body:

```
DEF twice(x) -> (y)
  t = ADD(x, x)
  y = ADD(t, x)
END

IN a
OUT = CALL twice(a)
```

The body is ordinary Jotnar. The formals are inputs to the subgraph. The declared outputs are the names the caller receives. The `END` closes the definition.

At the call site, every internal name is namespaced:

```
twice#1.t
twice#1.y
```

The suffix `#1` is the call-site number. A second call gets `#2`, and so on. This is how two calls to the same definition can coexist in one flat program without colliding. The namespacing is visible in diagnostics, which is useful when a macro expands into a dozen instructions.

CALL FORMS

All of these are accepted:

```
OUT = CALL twice(a)          # one output, named
x, y = CALL split(a)         # multiple outputs
x = CALL twice(a)            # one output, assigned
CALL twice(a)                # statement form; outputs are still bound
```

A definition cannot call itself, directly or indirectly. Recursion is a textual expansion loop, not a runtime stack, so the assembler rejects it:

```
recursive CALL: twice -> twice
```

LAYOUT CONTRACTS ON FORMALS

A formal may declare a layout:

```
DEF f(x AS F:SEQ) -> (y)
  y = x
END
```

The call site must satisfy it. If the actual argument is `T:SEQ`, the assembler reports a call-site layout violation. This is how a library can document the geometry it expects without relying on the caller's payload.

NO HIDDEN CONVERSIONS

A layout contract is not a cast. If the formal says `F:SEQ`, the caller must pass an `F:SEQ` stream. `T:SEQ` is not "close enough." Use `SQRT` or an explicit conversion if that is what you mean.

IMPORT

`IMPORT path` splices another listing into the current expansion. The path is resolved relative to the listing directory, with a `stdlib/` search path for bare names. `--stdlib DIR` overrides the search path.

```
IMPORT math.asm
```

Imports are textual and recursive. Cycles are detected and reported. Imported `IN` declarations stay `UNKNOWN` for layout purposes in the caller, matching the Python frontend. If you need a layout contract, put it on the `DEF` formal.

REPEAT AND BREAK

`REPEAT N` opens a block; `END` closes it. There are two behaviors.

Static unrolling. If `N` is an integer literal and the body contains no `BREAK`, the assembler copies the body `N` times. Names are not renamed between copies, so a reassignment in copy 2 replaces the name from copy 1. That is deliberate: it is how a sequence of updates is written.

```
IN x
REPEAT 3
  x = IADD(x, 1)
END
OUT = IADD(x, 0)
```

Runtime loop. If the body contains a `BREAK`, or if `N` is a stream name rather than a literal, the assembler keeps a loop marker and the interpreter iterates. `BREAK cond` exits the innermost loop when the first element of `cond` is nonzero.

```
IN x
IN limit
REPEAT limit
  x = IADD(x, 1)
  BREAK ISUB(x, limit)
END
```

`BREAK` outside a `REPEAT` is an error. `END` without `REPEAT` is an error. A nested loop must be closed before its parent.

> **STATIC VS. RUNTIME**
>
> Static unrolling is a compile-time convenience. A runtime loop is a real loop in the interpreter and in emitters that support it. If you need a deterministic instruction count, prefer a literal `REPEAT` without `BREAK`.

STATE AND RANGE

`STATE` marks a stream as carried across host iterations. It must also be declared `IN` (the seed) and assigned as an output (the carry). The CLI's `jotnar repeat` command threads the state from one iteration to the next.

```
IN x
STATE x
REPEAT 3
  x = IADD(x, 1)
END
OUT = IADD(x, 0)
```

`jotnar repeat` expects a payload of the form `{"iterations":[{...}, {...}]}`. Non-state inputs are read from each iteration; state streams seed from iteration 0 and carry forward.

`RANGE name lo hi` is parsed and bounds-checked. It is an annotation for host planners and readers; the reference interpreter does not emit an instruction for it.

CONFIG

`CONFIG key value` sets a program-wide constant. The registry reads these through typed helpers. The important ones:

KEY	USED BY	DEFAULT
m_acc	accumulation scale	31344
m_cov	coverage/output scale	33280
beta	BETA, BETA_V5, attention mask	0.5
beta_b	TBETA	falls back to beta
eps_rms / eps_rms_c	RMSNORM	4514 counts
eps_ln / eps_ln_c	LAYERNORM	4514 counts
rope_base	ROTARY, ROPE	10000.0

`CONFIG` is program-global and must appear outside `DEF`. It is not a runtime knob that changes per invocation; it is part of the program's arithmetic contract.

THE DIRECTIVE TABLE

DIRECTIVE	SCOPE	EFFECT
IN name [AS layout]	global	Declares an input stream.
OUT = ...	any	Names an output stream.
DEF name(args) -> (outs)	global	Defines a reusable subgraph.
CALL name(args)	any	Expands a definition at the call site.

END	block	Closes DEF or REPEAT.
IMPORT path	global	Splices another listing.
REPEAT N	any	Unrolls or loops a body.
BREAK cond	loop	Exits the innermost loop.
STATE name	global	Marks a stream as host-carried.
RANGE name lo hi	global	Annotates a numeric range.
CONFIG key value	global	Sets a program-wide constant.
# comment	any	Ignored to end of line.

DEF NAMES A GRAPH. CALL PASTES A FRESH COPY. REPEAT COPIES OR LOOPS. NOTHING IS HIDDEN, AND NOTHING IS RECURSIVE.

COMPOSITION RULE

END

THE DETERMINISTIC PRESS

THE INSTRUCTION SET

Eighty-six mnemonics, five families, one fail-loud registry.
Exact in the middle, float at the edge.

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06 EXACT INTEGERS

The integer family is the floor of the language. It wraps, it divides, it compares, and it never guesses.

Jotnar's integer streams are signed 64-bit values. Addition, subtraction, and multiplication wrap modulo 2^64. That is not an accident or a platform detail; it is the contract. If you need saturation, use `SADD`. If you need unsigned division, use `IDIVU`. If you need a different width, mask with `IAND`.

THE CORE

MNEMONIC	SIGNATURE	BEHAVIOR
IADD	I:*, * -> I:*	Wrapping add.
ISUB	I:*, * -> I:*	Wrapping subtract.
IMUL	I:*, * -> I:*	Wrapping multiply.
IMAD	I:*, *, * -> I:*	a*b + c, wrapping.
SADD	I:*, * -> I:*	Saturating add.
IMAX	I:*, * -> I:*	Elementwise maximum.
IDIV	I:*, * -> I:*	Truncating signed division.
IMOD	I:*, * -> I:*	Remainder of truncating division.
IDIVU	I:*, * -> I:*	Unsigned 64-bit division.
IMODU	I:*, * -> I:*	Unsigned 64-bit remainder.

`IDIV` and `IMOD` follow C's truncation toward zero, not Python's floor division. They raise on division by zero and on `INT64_MIN / -1`, which is unrepresentable. `IDIVU` and `IMODU` reinterpret the operands as unsigned 64-bit values before dividing.

BITS AND SHIFTS

MNEMONIC	SIGNATURE	BEHAVIOR
IAND	I:*, * -> I:*	Bitwise and.

IOR	I:*, * -> I:*	Bitwise or.
IXOR	I:*, * -> I:*	Bitwise exclusive-or.
ISHL	I:*, F:SCALAR -> I:*	Logical left shift by 0..63.
ISHR	I:*, F:SCALAR -> I:*	Logical right shift by 0..63.
ISAR	I:*, * -> I:*	Arithmetic right shift by 0..63.

`ISHL` and `ISHR` take a scalar shift amount. `ISAR` takes a stream or scalar. All three reject shift counts outside `0..=63`. There is no "shift by 64 means zero" rule; that is a bug waiting to happen.

SHIFT RANGE

A shift count of 64 is an error, not a zero. A negative count is an error. If you are generating shifts from data, clamp them first.

COMPARISONS AND SELECT

MNEMONIC	SIGNATURE	BEHAVIOR
IEQ	I:*, * -> I:*	1 if equal, else 0.
ILT	I:*, * -> I:*	1 if a < b, else 0.
ISELECT	I:*, *, * -> I:*	mask != 0 ? a : b.
F2I	F:* -> I:*	Float to int, truncating toward zero.
I2F	I:* -> F:*	Integer to F64.

`ISELECT` is the branch-like primitive. It is not a branch; both arms are computed, and the mask chooses elementwise. That makes it safe for data-dependent control flow in a graph and easy to lower to hardware.

```
# A two-way select. One condition, two data inputs, no arithmetic.
IN C
IN A
IN B

OUT = ISELECT(C, A, B)
```

`F2I` is strict. It rejects `NaN`, `Inf`, values outside the int64 range, and triples. It truncates toward zero, so `-3.9` becomes `-3`. If you need rounding, do it explicitly before the conversion.

INTEGERS AND JSON

The payload boundary accepts integers as JSON numbers or strings. Use strings for values above 2^53. The output side writes integers as strings for the same reason. The interpreter itself always uses int64; the string form is only a transport detail.

A WORKED EXAMPLE: WRAPPING

```
IN a
IN b
s = IADD(a, b)
t = IMUL(s, 3)
OUT = IAND(t, 255)
```

This is a wrapping add, a wrapping multiply by three, and an 8-bit mask. If `a` and `b` are bytes, `s` can carry into bit 8 and above; `IAND` keeps the low byte. That is the exact idiom used by the CPU8 listing in Chapter 16.

IF YOU WANT A DIFFERENT WIDTH, MASK IT. IF YOU WANT SATURATION, ASK FOR IT. IF YOU WANT A CONVERSION, NAME IT.

INTEGER RULE

WHY NOT JUST USE FLOATS?

Because floats are not a superset of integers. An int64 above 2^53 cannot be represented exactly in an F64. A checksum, an address, a byte count, or a hash is not a float. Jotnar keeps the integer path exact all the way to the emitter: the C target uses `uint64_t` arithmetic for wrapping ops, and the non-FPU target never touches a float for integer work.

The integer family is also the substrate for the CPU and byte chapters. Once you can add, mask, shift, compare, and select, you can build a machine.

END

07 TRIPLES AND THE PHI LATTICE

Float at the edge; exact in the middle. A triple is a value that carries its own scale.

A **triple** is three planes with the same shape:

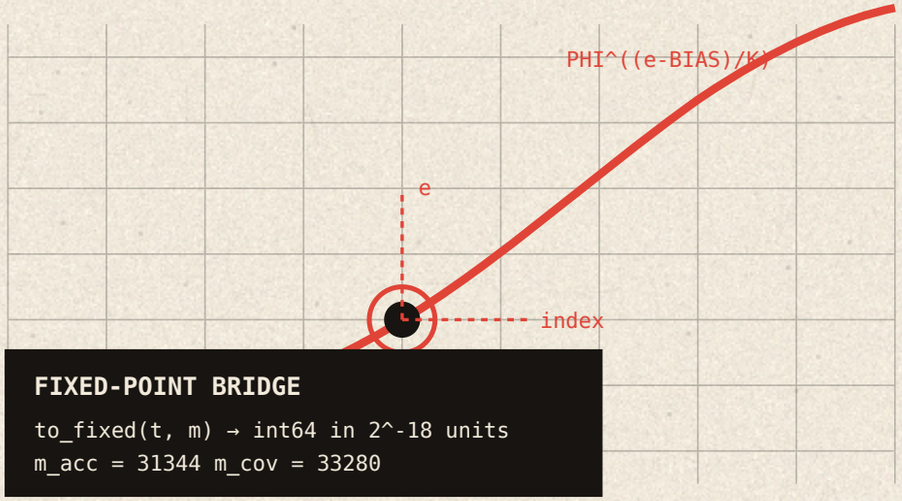
```
 s : sign       int8, -1 or +1
e : exponent   int32, biased by 32768
z : zero flag  uint8, 1 for exact zero
```

The value is reconstructed as:

```
value = s * PHI^((e - 32768) / 512)
```

where `PHI = 1.618033988749895` and the exponent step is `1/512` of a phi-log unit. The zero flag is separate because zero has no sign and no exponent; trying to encode it as one would lose the distinction between “zero” and “very small.”

THE PHI LATTICE: sign × PHI^((e - 32768) / 512) FIG. 03



TRIPLE

s : sign {-1, +1}

e : exponent (i32)

z : zero flag (u8)

same shape on all planes
no float in the middle

The phi lattice. Values are scale-free; the fixed-point bridge is where scale re-enters.

WHY THIS EXISTS

Floating point has a fixed number of significand bits. Every operation rounds. For a pipeline that composes dozens of operations, the rounding order matters, and different SIMD widths can produce different last bits. Jotnar's triple path is designed to make the arithmetic **integer-only in the middle** and to push float to the sensor boundary and the display boundary.

The lattice is not a floating-point format. It is a logarithmic encoding with a shared, frozen set of lookup tables. Multiplication and division become exponent adds and subtracts; addition and subtraction go through a fixed-point bridge. The result is deterministic and cheap to implement in integer hardware.

NOT A FLOAT

`T` is not a float kind. You cannot pass a triple to `F2I`; the operation rejects it. You cannot add a triple to a float with `ADD` unless the layouts agree. To leave the lattice, use `decode` at the boundary; to enter it, use `encode`.

THE FIXED-POINT BRIDGE

Most triple operations do not work directly on `(s,e,z)` pairs. They convert to a fixed-point integer at a chosen scale `m`, do integer arithmetic, and convert back. The bridge is:

```
to_fixed(t, m)   -> int64 in 2^-18 units
from_fixed(q, m) -> Triple
```

The scale `m` is a bias value, not a floating-point exponent. Two scales matter:

CONFIG	DEFAULT	USED FOR
m_acc	31344	accumulation, convolution, matmul
m_cov	33280	coverage, output, most elementwise ops

Set them with `CONFIG`:

```
CONFIG m_acc 41200
CONFIG m_cov 41200
```

The bridge uses frozen LUTs for the fractional part and a fine LUT for the exponent reconstruction. Rounding is round-half-to-even, matching numpy. The emitters bake the same tables, which is why the non-FPU target can reproduce the triple path without a single `float` or `double`.

SCALE IS PART OF THE PROGRAM

Changing `m_acc` or `m_cov` changes the arithmetic. It is not a display setting. If you change the scale, re-run the parity gate. The frozen defaults are part of the reference contract.

ARITHMETIC

MNEMONIC	SIGNATURE	BEHAVIOR
ADD	$A, $A -> $A	Elementwise add through the fixed-point bridge.
SUB	$A, $A -> $A	Elementwise subtract.
MUL	$A, $A -> $A	Elementwise multiply.
DIV	$A, $A -> $A	Elementwise divide.
SQUARE	$A -> $A	Elementwise square.

SQRT	F:HW -> T:HW	Square root, emitted as triples.
SIGMOID	$A -> $A	Logistic sigmoid through the sigmoid LUT.
CLIP	$A, F:SCALAR, F:SCALAR -> $A	Clamp to [lo, hi].
RESCALE	$A, F:SCALAR -> $A	Change the fixed-point scale.
GT	$A, $A -> I:*	Elementwise greater-than, returning a mask.
SELECT	I:*, $A, $A -> $A	Masked select.

`MUL` and `DIV` are not IEEE operations. They are lattice operations: exponents combine, signs multiply, and the zero flag dominates. That is why the result can be exact in a way a float multiply is not.

A WORKED EXAMPLE

```
CONFIG m_cov 33280
IN a AS T:HW
IN b AS T:HW
IN mask AS I:HW

sum  = ADD(a, b)
sq   = SQUARE(a)
gate = GT(a, b)
out  = SELECT(mask, sq, sum)
OUT  = CLIP(out, 0.0, 1.0)
```

Every step is elementwise. `GT` produces an integer mask; `SELECT` chooses between two triple streams; `CLIP` stays in the lattice. No float appears in the listing. The emitters can lower this to integer-only C, CUDA FP32, or hardware, depending on the target.

FLOAT AT THE EDGE; EXACT IN THE MIDDLE. THE TRIPLE IS HOW THE MIDDLE STAYS EXACT.

LATTICE RULE

READING A TRIPLE STREAM

A triple stream is three arrays of the same shape. In JSON:

```
{"kind":"T","shape":[2,2],
 "s":[1,-1,1,1],
 "e":[30000,31000,32000,33000],
 "z":[0,0,0,0]}
```

The first element is positive with exponent 30000; the second is negative with exponent 31000; and so on. All four are non-zero. If `z` were 1, the value would be exactly zero regardless of `s` and `e`.

That is the whole representation. It is small, it is integer, and it is the same on every machine that Jotnar targets.

END

08 LINEAR ALGEBRA AND THE TRANSFORMER

Matrix multiply, norms, softmax, attention, and the small set of moves that make a transformer a graph.

The linear-algebra family is where the triple path earns its keep. A matmul is a long chain of multiplies and adds; if each one rounded independently, the result would depend on the order of accumulation. Jotnar accumulates in fixed-point integers at `m_acc`, then returns to the lattice at `m_cov`. The order is explicit, and the emitters preserve it.

MATMUL

MNEMONIC	SIGNATURE	BEHAVIOR
MATMUL	*, * -> *	2D matrix multiply over triples.
BATCH_MATMUL	*, * -> *	Batched matmul over leading dimensions.
BMMV	*13 -> *	Fused batched matrix-vector kernel with explicit strides.

The runtime checks the inner dimensions:

```
a.shape[-1] == b.shape[-2]
```

The classic mistake is a missing transpose. Jotnar names it:

```
inner dims [B,M,K] vs [B,N,K] (did you forget TRANSPOSE?)
```

A 2×2 multiply is small enough to read:

MATMUL2 / SHOWCASE/LANGUAGE/MATMUL2.ASM

```
# 2x2 matrix multiply. Four dot products with a shared operand graph.
IN a00
IN a01
IN a10
IN a11
IN b00
IN b01
IN b10
IN b11

t00 = IMUL(a00, b00)
t01 = IMUL(a01, b10)
c00 = IADD(t00, t01)
t10 = IMUL(a10, b00)
t11 = IMUL(a11, b10)
c10 = IADD(t10, t11)
t02 = IMUL(a00, b01)
t03 = IMUL(a01, b11)
c01 = IADD(t02, t03)
t12 = IMUL(a10, b01)
t13 = IMUL(a11, b11)
c11 = IADD(t12, t13)
```

The listing is a graph: `t00` and `t01` feed `c00`; `t10` and `t11` feed `c10`; and the second column reuses `a00`, `a01`, `a10`, `a11`. The order is chosen for reading, not for a machine's pipeline. An emitter is free to schedule the independent products in any order that preserves the dataflow.

NORMS AND ACTIVATIONS

MNEMONIC	SIGNATURE	BEHAVIOR
RMSNORM	$X, $W -> $X	RMSNorm over the last axis.
LAYERNORM	$X, $W, $W -> $X	LayerNorm over the last axis; 2D input.
SILU	$A -> $A	x * sigmoid(x).
GELU	$A -> $A	GELU through frozen LUTs.
SIGMOID	$A -> $A	Logistic sigmoid.
SWIGLU	$A, F:SCALAR -> $A	silu(gate) * up; the input is 2*inter wide.
SCAN	$X, $X, $X -> $X	Recurrence step h*abar + bx.

`RMSNORM` and `LAYERNORM` take their epsilon from `CONFIG` (`eps_rms`, `eps_ln`) or from explicit count values (`eps_rms_c`, `eps_ln_c`). The default is 4514 counts at the coverage scale. That is a deliberate, frozen calibration, not a magic float.

SOFTMAX AND ATTENTION

MNEMONIC	SIGNATURE	BEHAVIOR
SOFTMAX	$A -> $A	Softmax over the last axis at BIAS scale.
SOFTMAX_WIDE	$A -> $A	Softmax with the wide fixed-point range.
TSHIFT	$A -> $A	Subtract the per-row maximum.
ATTN	8 args -> $A	Multi-head attention with a mask.
ROTARY	$X, * -> $X	Rotary embedding over triple pairs.
ROPE	$A, *, F:SCALAR, F:SCALAR -> $A	Float RoPE over (rows, heads*head_dim).
ARGMAX	$A, F:SCALAR -> I:*	Index of the maximum along an axis.
TOPK	$A, F:SCALAR -> $A, I:*	Top-k values and indices.

`SOFTMAX` requires BIAS-scale counts. If you feed it a stream at a different scale, it refuses. `TSHIFT` is the numerical-stability step: subtract the row maximum before exponentiating. The exp itself uses a frozen LUT, not `exp()` from libm.

`ATTN` takes query, key, value, a mask, and four integers: number of heads, number of KV heads, head dimension, and sequence length. The mask is `(per, S)`, where `per = nh / nkv`. It applies the mask by writing a large negative value into the scores, not by branching.

ROPE VS. ROTARY

`ROPE` is the float path: input `(rows, heads*head_dim)`, positions, head count, head dimension. `ROTARY` is the triple path: input triples and a position stream. They implement the same rotation idea at different stages of the pipeline. Use the one that matches your kind.

ANATOMY OF A BLOCK

A transformer block is not a single mnemonic. It is a graph:

1. `RMSNORM` or `LAYERNORM` on the input.
2. `MATMUL` with the Q, K, V weights.
3. `ROTARY` or `ROPE` on Q and K.
4. `ATTN` with the mask.

5. `MATMUL` with the output projection.
6. Residual `ADD`.
7. Norm, then `SWIGLU` or `GELU`.
8. Down projection and residual `ADD`.

Every step is an ordinary stream operation. There is no “transformer” op. That is the point: the model is a graph, and the graph is visible.

THE LISTING FIXES THE DATAFLOW, NOT THE SCHEDULE. THE EMITTER MAY REORDER INDEPENDENT WORK; IT MAY NOT CHANGE THE ACCUMULATION ORDER INSIDE A MATMUL.

SCHEDULING RULE

MOVES AROUND THE MATH

The linalg path depends on a small set of shape moves:

MNEMONIC	SIGNATURE	BEHAVIOR
TRANSPOSE	$A -> $A^T	Swap the last two dimensions.
RESHAPE2	*, F:SCALAR, F:SCALAR -> *	Reshape to two dimensions.
RESHAPE3	*, F:SCALAR, F:SCALAR, F:SCALAR -> *	Reshape to three dimensions.
PERMUTE3	*, F:SCALAR, F:SCALAR, F:SCALAR -> *	Permute three dimensions.
CONCAT	$A, $A, F:SCALAR -> $A	Concatenate along an axis.
SLICE	$A, F:SCALAR, F:SCALAR, F:SCALAR -> $A	Slice along an axis.
GATHER	*, I:* -> *	Gather rows by index.
SCATTER	$A, $A, I:* -> $A	Scatter values into a copy.

These are not cosmetic. A reshape changes the stride interpretation; a transpose changes the layout; a concat changes the axis length. The runtime checks what it can, and the emitters infer static shapes before lowering. If a shape is dynamic where the C emitter needs a constant, the emitter says so instead of guessing.

END

09 IMAGES, FIELDS, AND VISION

An image is a stream with channels. A field is a stream with a geometry. The image family is where the two meet.

The image family works on `H×W` and `H×W×C` streams. The usual pipeline is:

```
 U8:HWC  --SRGB_DECODE-->  F:HWC  --LUMA-->  F:HW
F:HW     --SQRT-->         T:HW   --SPLAT_BLUR-->  T:HW + T:HW
T:HW     --BETA-->         T:HW   --GAIN-->        F:HWC
F:HWC    --SRGB_ENCODE-->  U8:HWC
```

Every arrow is a stream. Every stage can be inspected, replaced, or run on a different target. There is no hidden image object.

COLOR AND LUMA

MNEMONIC	SIGNATURE	BEHAVIOR
SRGB_DECODE	U8:HWC -> F:HWC	Normalize bytes to [0,1], then pow(2.2).
SRGB_ENCODE	F:HWC -> U8:HWC	pow(1/2.2), scale to 0..255, clamp.
LUMA	F:HWC -> F:HW	Rec.709 weights: 0.2126, 0.7152, 0.0722.
GAIN	F:HWC, F:HW, T:HW -> F:HWC	Coherence-gated gain.

`sRGB` is not a linear-light space. Decoding before filtering and encoding after is the difference between a blur that looks right and a blur that looks muddy. The gamma is a fixed 2.2 approximation, not the piecewise sRGB curve; that is the frozen contract.

FILTERING

MNEMONIC	SIGNATURE	BEHAVIOR
GAUSS	$A, F:SCALAR, F:SCALAR -> $A	Gaussian blur with radius and sigma.
ISO_BLUR	T:HW -> T:HW	Fixed 5×5 Gaussian (radius 2, sigma 1).
CONV	*, * -> *	2D convolution with an explicit kernel.

DECONV	*, *, F:SCALAR, F:SCALAR -> *	Transposed convolution with stride and padding.
INTERP	*, F:SCALAR, F:SCALAR -> *	Bilinear resize by dyadic scales.
POOLAVG	* -> *	Global average pool over H,W; returns (C).
PRELU	$A, $A -> $A	PReLU with scalar or per-channel slope.

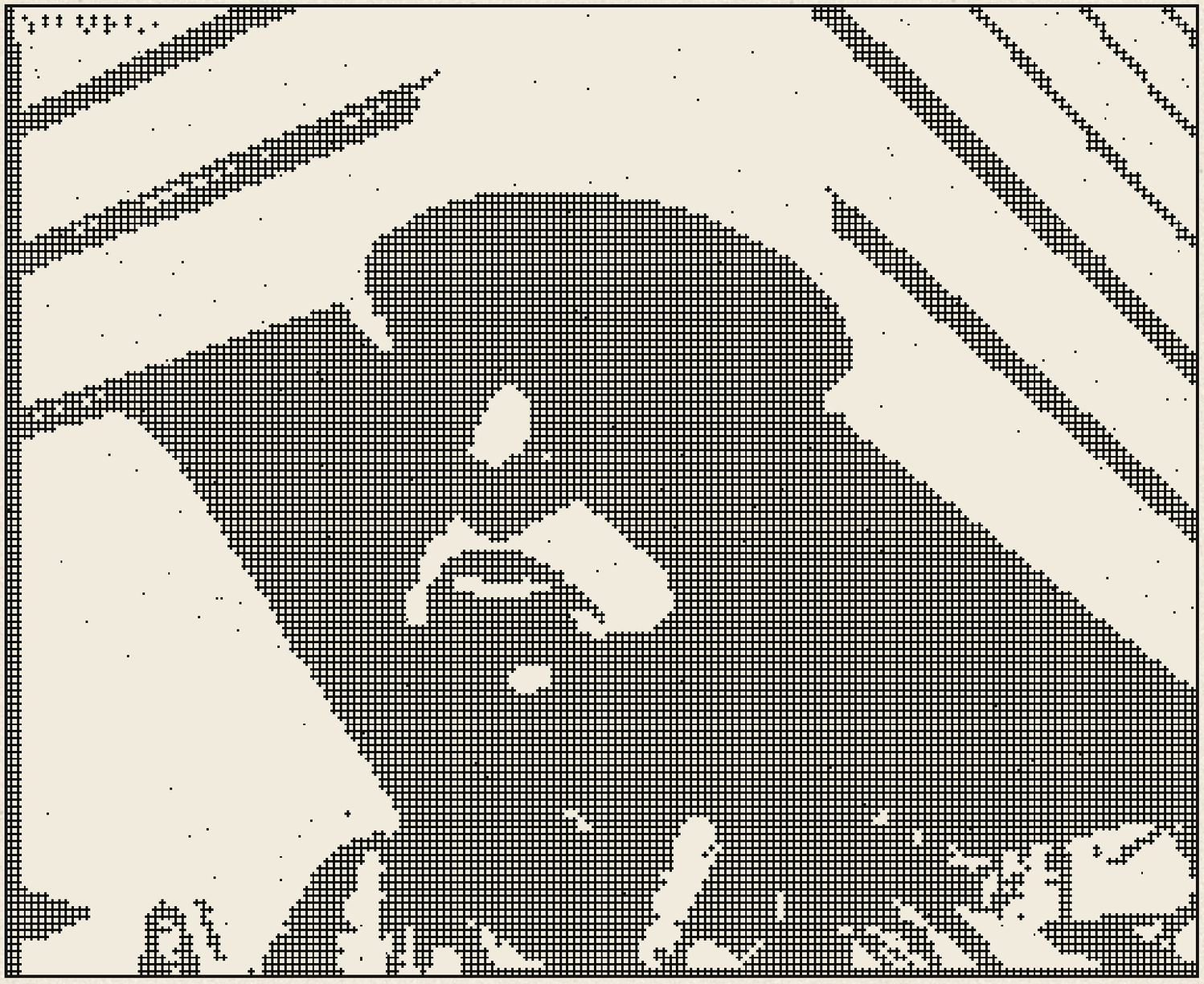
`INTERP` only accepts dyadic scales (powers of two). A non-dyadic scale is a lowering error, not a silent approximation. That is a deliberate constraint: it keeps the fixed-point coordinate math exact.

STRUCTURE AND COHERENCE

`SPLAT_BLUR` is the big one. It computes a structure tensor (Sobel gradients, then a Gaussian window), derives coherence, buckets the local orientation, and blends several oriented splats. It returns two streams: the filtered image and the coherence field.

MNEMONIC	SIGNATURE	BEHAVIOR
SPLAT_BLUR	T:HW -> T:HW, T:HW	Structure-tensor splat blur; image + coherence.
BETA	$A -> $A	Fill with the scalar beta.
BETA_V5	T:HW, T:HW -> T:HW	Beta field v5 from depth and coherence.
MIXDYAD	T:HW, T:HW, I:HW -> T:HW	Blend current and history under a mask.
WARP	T:HW, F:HW2 -> T:HW	Bilinear warp by a flow field.
STATIC	F:HW2 -> I:HW	Mask true where flow magnitude is zero.

Coherence is a number in `[0,1]`: how strongly the local gradient points in one direction. The beta field uses it to decide how much to trust the current estimate versus the history. `BETA_V5` adds a depth term. `MIXDYAD` blends under a mask, and `WARP` resamples the previous field by a flow.



A face through the image path, reduced to a halftone. The original is a 360×288 photograph; the portrait is not the point, the pipeline is. Image: Jotnar samples/input_example.png.

THE 43-OP DEMO

The image demo runs one 43-op geometric listing through C, non-FPU, and CUDA. It is the shortest way to see the same graph on three machines:

```
cargo run --release --bin jotnar-demo-img -- \
  --input samples/input_example.png --gallery gallery
```

The gallery contains the C output, the non-FPU output, and the CUDA output. The point is not that the images are pixel-identical across every target; floats can differ in the last ulp. The point is that the graph is the same, the shapes are the same, and the exact moves are exact.

> **WHAT "COHERENCE" IS NOT**
>
> Coherence is not confidence. It is a local orientation measure. A high-coherence edge is a direction, not a truth claim. The beta field uses it as one input among several; it does not turn it into a probability.

WHY IMAGES BELONG IN AN ASSEMBLY MANUAL

Because the image path is where the language's constraints become visible. You cannot blur a `U8:HWC` stream directly; you decode first. You cannot resize by an arbitrary scale; you use a dyadic scale or you get an error. You cannot treat a triple as a float; you decode at the boundary. The image family is a stress test for the type system, and it passes by refusing the wrong graph. The field-note version of the 43-op run is Chapter 17.

DECODE AT THE EDGE, FILTER IN THE MIDDLE, ENCODE AT THE EDGE. THE MIDDLE IS WHERE THE EXACTNESS LIVES.

IMAGE RULE

END

10 BYTES, CONTAINERS, AND MASKS

A byte stream is an integer stream with a small range. Everything else is indexing, checking, and refusing to read past the end.

The byte family treats a 1D integer stream as a sequence of bytes. There is no separate `bytes` value at the language level; a byte is an `I64` in `[0,255]` (or a `U8` at the payload boundary). The operations are explicit about bounds, because a decoder that silently reads past the end is a security bug with a nicer name.

THE BYTE OPERATIONS

MNEMONIC	SIGNATURE	BEHAVIOR
IWINDOW	I:*, *, F:SCALAR -> I:*	Slice [start, start+len).
IFIND	I:*, I:* -> I:*	First index of a pattern; -1 if absent.
IGATHER1	I:*, I:* -> I:*	Gather from a 1D table by indices.
IASSEMBLE	I:*, F:SCALAR, F:SCALAR -> I:*	Assemble 1..8 little-endian bytes into an integer.
ISUM	I:*, F:SCALAR, F:SCALAR -> I:*	Sum a slice.
ISCATTER1	I:*, I:*, I:* -> I:*	Scatter values into a 1D copy at indices.
ICRC32	I:* -> I:*	CRC-32 (IEEE) over bytes.
ICRC32C	I:*, I:* -> I:*	CRC-32C with an explicit seed.
IVARINT	I:*, * -> I:*, I:*	Decode a varint; returns value and bytes consumed.
IINFLATE	I:*, F:SCALAR -> I:*	Raw DEFLATE decode with an expected size.

Every slice checks `start >= 0`, `len >= 0`, and `start + len <= len(x)`. Every gather checks each index against the table length. Every scatter checks each index against the destination. `IVARINT` reports truncation; `IINFLATE` reports a bad stream; `IFIND` returns `-1` rather than throwing, because “not found” is a normal result.

BOUNDS ARE NOT OPTIONAL

`IWINDOW(x, -1, 4)` is an error. `IGATHER1(table, [999])` is an error. `ISCATTER1` with an out-of-range index is an error. The runtime does not clamp, wrap, or skip. If you need a safer index, compute it with `IAND` or `IMOD` and check it first.

CHECKSUMS

`ICRC32` computes the IEEE CRC-32 of a byte stream, with the usual initial value and final XOR:

```
IN bytes
OUT = ICRC32(bytes)
```

`ICRC32C` takes an explicit seed, so a protocol can continue a checksum across chunks:

```
IN chunk
IN seed
OUT = ICRC32C(chunk, seed)
```

The polynomial is part of the mnemonic, not a configuration knob. If you need a different CRC, write it with the integer family; the byte family stays small on purpose.

VARINTS AND INFLATE

`IVARINT` decodes a base-128 varint at a start index and returns two streams: the value and the number of bytes consumed. That second output is what makes it composable: you can advance the cursor without guessing how long the encoding was.

```
IN data
value, used = IVARINT(data, 0)
next = IADD(used, 0)
```

`IINFLATE` decodes a raw DEFLATE stream with an expected output size. The expected size is a scalar, not a buffer size hint; the decoder allocates exactly that many bytes and fails if the stream would produce a different count. That keeps the operation deterministic and avoids a class of decompression bombs.

CONTAINERS

The container operations are shape moves, not byte moves:

MNEMONIC	SIGNATURE	BEHAVIOR
GATHER	*, I:* -> *	Gather rows by index.
GATHER3	*, I:*, I:* -> *	Gather from a 3D table by i and j.
SCATTER	$A, $A, I:* -> $A	Scatter values into a copy.
SLICE	$A, F:SCALAR, F:SCALAR, F:SCALAR -> $A	Slice along an axis.
CONCAT	$A, $A, F:SCALAR -> $A	Concatenate along an axis.
SELECT	I:*, $A, $A -> $A	Masked select.
GT	$A, $A -> I:*	Greater-than mask.

`GATHER` checks that the index vector is in range for the first axis. `GATHER3` checks that `i` and `j` are 1D, the same length, and in range for the table's first two axes. `SCATTER` writes into a copy; it does not mutate the input stream.

WHY A COPY?

The language has no mutation. A "scatter" is a new stream with some positions overwritten. That is what makes it safe to reorder independent work: no two operations can race on the same underlying buffer because there is no shared mutable buffer in the graph.

TWO REAL CONSUMERS

The Futhark demo uses `IWINDOW` to slide a four-character window over a word, `IXOR` / `IMUL` / `IADD` to compute a fingerprint, and `IGATHER1` to read a sorted lexicon table. The DOOM runner uses the same byte family for a compiled RV32IM image: fetch bytes, assemble instructions, decode fields, and checksum frames.

Neither demo needed a new byte type. The integer family plus bounds checks was enough.

A BYTE IS AN INTEGER WITH A SMALL RANGE. A DECODER IS A GRAPH WITH A BOUNDS CHECK. NEVER CONFUSE THE TWO.

BYTE RULE

END

11 THE COMPLETE REGISTER

The registry is the authority. Every mnemonic has an arity, a signature, a family, and a reference implementation. If it is not here, it is not in the language.

This is the full register. It is generated from the same data the manual uses for the per-family chapters, and it is checked against the Rust registry's 86 entries.

Read the columns as follows:

- **Mnemonic** — the exact name the assembler binds. Names are uppercase.
- **In→Out** — the arity. A `*` in a signature means "any layout."
- **Signature** — the layout pattern. `$A` is a unified variable; `*` binds nothing; `KIND:GEOM` is a literal.
- **What it does** — the contract in one line. The source and the parity harness are the long version.

THE REGISTER IS NOT A SUGGESTION

An unknown mnemonic is an error. A wrong arity is an error. A layout conflict is an error. A registered mnemonic with no pattern for a target is a loud `NoPattern`, not a silent fallback. The table is a map, not a permission slip.

MNEMONIC	IN → OUT	SIGNATURE	WHAT IT DOES
IADD	2→1	I:*, * -> I:*	Wrapping signed 64-bit addition.
SADD	2→1	I:*, * -> I:*	Saturating signed 64-bit addition.
IMAX	2→1	I:*, * -> I:*	Elementwise maximum.
ISUB	2→1	I:*, * -> I:*	Wrapping signed 64-bit subtraction.
IMUL	2→1	I:*, * -> I:*	Wrapping signed 64-bit multiplication.
IMAD	3→1	I:*, *, * -> I:*	Fused multiply-add: a*b + c, wrapping.
IDIV	2→1	I:*, * -> I:*	Truncating signed division; zero and MIN/-1 are errors.
IMOD	2→1	I:*, * -> I:*	Remainder from truncating signed division.
IDIVU	2→1	I:*, * -> I:*	Unsigned 64-bit division.
IMODU	2→1	I:*, * -> I:*	Unsigned 64-bit remainder.
IXOR	2→1	I:*, * -> I:*	Bitwise exclusive-or.
IAND	2→1	I:*, * -> I:*	Bitwise and.

IOR	2→1	I:*, * -> I:*	Bitwise or.
ISAR	2→1	I:*, * -> I:*	Arithmetic right shift by 0..63.
IEQ	2→1	I:*, * -> I:*	1 when equal, else 0.
ILT	2→1	I:*, * -> I:*	1 when a < b, else 0.
ISELECT	3→1	I:*, *, * -> I:*	Masked select: mask != 0 ? a : b.
ISHL	2→1	I:*, F:SCALAR -> I:*	Logical left shift by a scalar 0..63.
ISHR	2→1	I:*, F:SCALAR -> I:*	Logical right shift by a scalar 0..63.
F2I	1→1	F:* -> I:*	Float to integer, truncating toward zero.
I2F	1→1	I:* -> F:*	Integer to F64.

MNEMONIC	IN → OUT	SIGNATURE	WHAT IT DOES
MATMUL	2→1	*, * -> *	2D matrix multiply with fixed-point accumulation.
BATCH_MATMUL	2→1	*, * -> *	Batched matmul over leading batch dimensions.
SOFTMAX	1→1	$A -> $A	Softmax over the last axis at BIAS scale.
SOFTMAX_WIDE	1→1	$A -> $A	Softmax with the wide fixed-point range.
TSHIFT	1→1	$A -> $A	Subtract the per-row maximum (softmax stabilization).
RMSNORM	2→1	$X, $W -> $X	RMSNorm over the last axis with a weight vector.
LAYERNORM	3→1	$X, $W, $W -> $X	LayerNorm over the last axis (2D input).
SILU	1→1	$A -> $A	SiLU activation: x * sigmoid(x).
GELU	1→1	$A -> $A	GELU activation through frozen LUTs.
SCAN	3→1	$X, $X, $X -> $X	Recurrence step: h * abar + bx.
ROTARY	2→1	$X, * -> $X	Rotary embedding over triple pairs by position.
ROPE	4→1	$A, *, F:SCALAR, F:SCALAR -> $A	Float RoPE over (rows, heads * head_dim).
SWIGLU	2→1	$A, F:SCALAR -> $A	Gated MLP: silu(gate) * up.
ATTN	8→1	$A,$A,$A,I:*,F:SCALAR,F:SCALAR,F:SCALAR,F:SCALAR -> $A	Multi-head attention with a causal/validity mask.
BMMV	13→1	*13 -> *	Fused batched matrix-vector kernel with explicit strides.

MNEMONIC	IN → OUT	SIGNATURE	WHAT IT DOES
TRANSPOSE	1→1	$A -> $A^T	Swap the last two dimensions; layout becomes ^T.
GAUSS	3→1	$A, F:SCALAR, F:SCALAR -> $A	Gaussian blur with radius and sigma.
RESHAPE2	3→1	*, F:SCALAR, F:SCALAR -> *	Reshape to two dimensions.
RESHAPE3	4→1	*, F:SCALAR, F:SCALAR, F:SCALAR -> *	Reshape to three dimensions.
PERMUTE3	4→1	*, F:SCALAR, F:SCALAR, F:SCALAR -> *	Permute three dimensions.

SELECT	3→1	I:*, $A, $A -> $A	Elementwise select: mask != 0 ? a : b.
CONCAT	3→1	$A, $A, F:SCALAR -> $A	Concatenate two streams along an axis.
ARGMAX	2→1	$A, F:SCALAR -> I:*	Index of the maximum along an axis.
TOPK	2→2	$A, F:SCALAR -> $A, I:*	Top-k values and indices along the last axis.
GT	2→1	$A, $A -> I:*	Elementwise greater-than, returning an integer mask.
SLICE	4→1	$A, F:SCALAR, F:SCALAR, F:SCALAR -> $A	Slice along an axis from start to end.
CLIP	3→1	$A, F:SCALAR, F:SCALAR -> $A	Clamp every value into [lo, hi].
DIV	2→1	$A, $A -> $A	Elementwise division.
SIGMOID	1→1	$A -> $A	Logistic sigmoid.
RESCALE	2→1	$A, F:SCALAR -> $A	Rescale triples to a new fixed-point scale.
GATHER	2→1	*, I:* -> *	Gather rows by an index vector.
GATHER3	3→1	*, I:*, I:* -> *	Gather from a 3D table by i/j index vectors.
SCATTER	3→1	$A, $A, I:* -> $A	Scatter values into a copy at the given indices.
PRELU	2→1	$A, $A -> $A	PReLU with a scalar or per-channel slope.
POOLAVG	1→1	* -> *	Global average pool over H,W; returns (C).
DECONV	4→1	*, *, F:SCALAR, F:SCALAR -> *	Transposed convolution with stride and padding.
INTERP	3→1	*, F:SCALAR, F:SCALAR -> *	Bilinear resize by dyadic scales.
CONV	2→1	*, * -> *	2D convolution with an explicit kernel.

MNEMONIC	IN → OUT	SIGNATURE	WHAT IT DOES
SRGB_DECODE	1→1	U8:HWC -> F:HWC	Decode sRGB bytes to linear float (normalized, gamma 2.2).
LUMA	1→1	F:HWC -> F:HW	Rec.709 luma: 0.2126 R + 0.7152 G + 0.0722 B.
SQRT	1→1	F:HW -> T:HW	Elementwise square root, emitted as triples.
SPLAT_BLUR	1→2	T:HW -> T:HW, T:HW	Structure-tensor splat blur; returns image and coherence.
SUB	2→1	$A, $A -> $A	Elementwise subtraction.
BETA_V5	2→1	T:HW, T:HW -> T:HW	Beta field v5 from depth and coherence.
MUL	2→1	$A, $A -> $A	Elementwise multiplication.
ADD	2→1	$A, $A -> $A	Elementwise addition.
SQUARE	1→1	$A -> $A	Elementwise square.
GAIN	3→1	F:HWC, F:HW, T:HW -> F:HWC	Apply a coherence-gated color gain.
SRGB_ENCODE	1→1	F:HWC -> U8:HWC	Encode linear float to sRGB bytes.
BETA	1→1	$A -> $A	Fill a stream with the scalar beta, encoded as triples.
TBETA	1→1	$A -> $A	Fill a stream with beta_b (temporal beta).
ISO_BLUR	1→1	T:HW -> T:HW	Fixed 5x5 Gaussian blur (radius 2, sigma 1).

WARP	2→1	T:HW, F:HW2 -> T:HW	Bilinear warp of depth by a flow field; None passes through.
STATIC	1→1	F:HW2 -> I:HW	Mask true where the flow magnitude is zero.
MIXDYAD	3→1	T:HW, T:HW, I:HW -> T:HW	Blend current and history samples under a mask.

MNEMONIC	IN → OUT	SIGNATURE	WHAT IT DOES
IWINDOW	3→1	I:*, *, F:SCALAR -> I:*	Slice a 1D integer/byte stream [start, start+len).
ICRC32	1→1	I:* -> I:*	CRC-32 (IEEE) over a byte stream.
ICRC32C	2→1	I:*, I:* -> I:*	CRC-32C with an explicit seed.
IINFLATE	2→1	I:*, F:SCALAR -> I:*	Raw DEFLATE decode with an expected output size.
IVARINT	2→2	I:*, * -> I:*, I:*	Decode a varint; returns value and bytes consumed.
IFIND	2→1	I:*, I:* -> I:*	First occurrence of a pattern; -1 when absent.
IGATHER1	2→1	I:*, I:* -> I:*	Gather from a 1D table by indices.
IASSEMBLE	3→1	I:*, F:SCALAR, F:SCALAR -> I:*	Assemble 1..8 little-endian bytes into an integer.
ISUM	3→1	I:*, F:SCALAR, F:SCALAR -> I:*	Sum a slice of a 1D integer stream.
ISCATTER1	3→1	I:*, I:*, I:* -> I:*	Scatter values into a 1D copy at indices.

END

THE DETERMINISTIC PRESS

THE MACHINE

The same plan on a portable C target, an integer-only target, a GPU, a piece of hardware, and a 32-bit machine.

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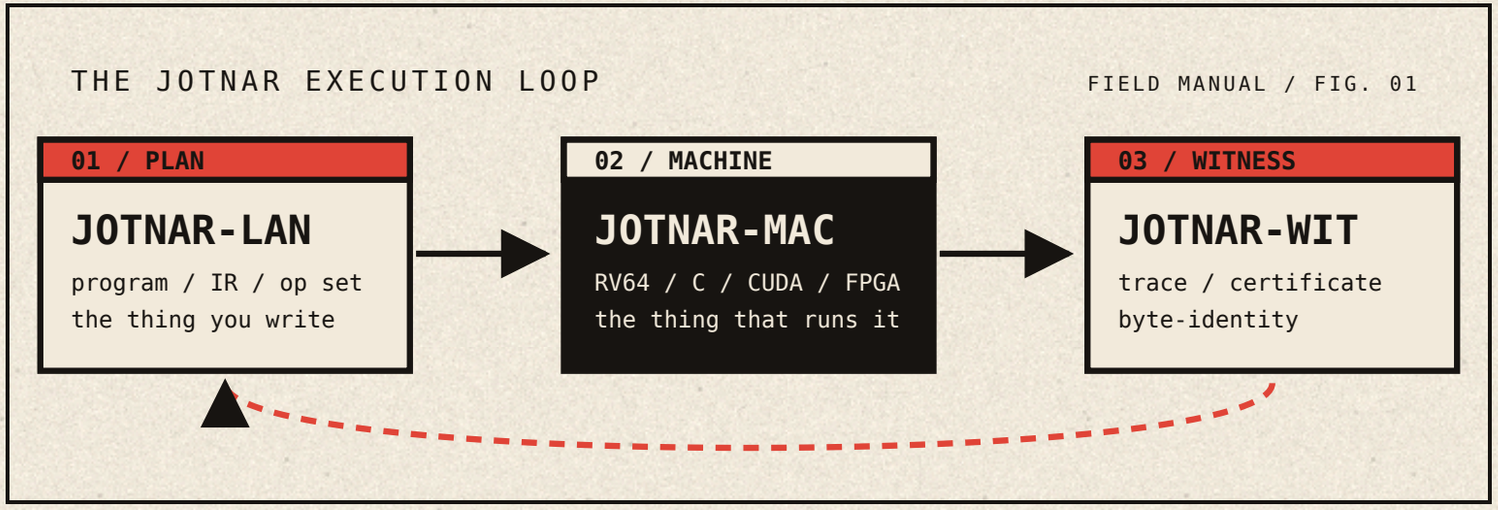
12 THE EXECUTION LOOP: LAN → MAC → WIT

A plan is not a machine. A machine is not a proof. The loop is what makes the difference visible.

Running a Jotnar program is a loop with three stations:

1. **Jotnar-LAN** — the program is parsed, expanded, bound, and (optionally) verified.
2. **Jotnar-MAC** — the bound graph is executed by the reference interpreter or lowered to a target.
3. **Jotnar-WIT** — the outputs are compared against an oracle, byte for byte where the contract allows it.

The stations are not a pipeline in the software-engineering sense. They are aspects of the same claim: the program you wrote, the machine that ran it, and the evidence that the result is what the plan said.



The execution loop. Same program + same input bytes = same output bytes. The witness is the third station, not an afterthought.

WHAT `jotnar run` DOES

The reference path is intentionally boring:

```
read listing
  -> expand IMPORT / DEF / REPEAT
  -> parse directives and assignments
  -> bind each mnemonic to the registry (arity + signature)
  -> read payload JSON
  -> bind input streams (kind + shape)
  -> execute the bound instructions in order
  -> re-check shapes at each op that has a rule
  -> write every stream (or the named outputs) as JSON
```

There is no JIT, no speculative execution, and no hidden caching. The interpreter walks the bound instruction list. `REPEAT` with a runtime count and `BREAK` are the only control flow; everything else is dataflow.

STATE AND REPEAT

`STATE` streams are seeded from the payload and carried across host iterations by `jotnar repeat`. Inside a single run, a `REPEAT` loop can update a name; the interpreter threads the new value into the next iteration. The state contract is checked at assembly time: a `STATE` name must also be an `IN` and must be assigned as an output.

THE ARTIFACTS

ARTIFACT	ASPECT	WHAT IT IS
A listing (.asm)	Jotnar-LAN	The program text.
A bound program	Jotnar-LAN	Parsed instructions plus registry signatures.
A target source (.c, .cu, .v)	Jotnar-MAC	The lowered machine program.
A payload (.json)	Jotnar-MAP	Typed input streams at rest.
An output map (.json)	Jotnar-MAP	Typed output streams.
A parity report	Jotnar-WIT	Comparisons against the oracle.
A trace / certificate	Jotnar-WIT	Evidence of execution, not a proof of intent.

The vocabulary matters because the claims are different. “It compiles” is a LAN claim. “It runs” is a MAC claim. “It matches the oracle” is a WIT claim. Do not use one to stand in for another.

WHAT THE WITNESS ACTUALLY CHECKS

The Rust port is gated by differential harnesses:

GATE	WHAT IT COMPARES	CURRENT RESULT
parity_ops.py	One small listing per mnemonic, Rust vs. Python.	88/88 op cases.
parity_emit.py	Emitted C and non-FPU binaries vs. Python-emitted binaries.	32 emitter cases.
parity_programs.py	The flagship programs, including the image listing.	Streams match within the stated class.
parity_cuda.py	CUDA units, bigram, argmax, headt/fp16/D32/D64.	Units within 1e-4; decisions bit-exact.
parity_futhark.py	The word-level Futhark listing vs. the JS rule engine.	118/118 words.
parity_doom_native.py	Frame 1/2 state and framebuffer vs. C and GPU cores.	Byte-identical where gated.

Triples and integers are compared bit-exactly. Floats are compared within a tight tolerance, because a different SIMD reduction order can change the last ulp. The manual says “exact” only where the gate is exact.

THE WITNESS RULE

The board is a witness, not a proof. Byte-identity against an oracle means the lowering preserved the plan under the tested inputs. It does not mean the plan is correct, the inputs are representative, or the model is honest. Those are separate claims. Appendix A turns this rule into a certificate schema.

A SMALL TRACE

A `verify` run is the cheapest witness:

```
$ jotnar verify tests/fixtures/cpu8.asm
layouts resolved: 53/59 (0 error(s))
```

That line says the declared layouts unify. It does not say the CPU executes correctly. For that, the Verilog backend emits the same integer program as structural hardware and checks the cycle trace against an independent ISA model. The trace is the witness; the 53/59 is a static check.

For a run, the witness is the output map:

```
{"streams":{"OUT":{"kind":"I64","shape":[4],
  "data":["13","26","39","52"]}}}
```

If the oracle says the same bytes, the run is witnessed. If it says something else, the run is a bug report.

“

PROGRAM, MACHINE, WITNESS. NAME THE ASPECT OR THE CLAIM IS AMBIGUOUS.

LOOP RULE

END

13 EMITTERS: C, NON-FPU, CUDA, VERILOG

The same bound graph can become portable C, integer-only C, a GPU kernel, a piece of hardware, or a 32-bit machine. The lowering is where the promises are tested.

An emitter consumes the bound program and produces a target artifact. It does not re-parse the source, and it does not get to reinterpret the graph. The registry's semantics are the contract; the emitter's job is to preserve them.

THE TARGETS

TARGET	FLAG	WHAT IT EMITS	WHAT IT REFUSES
C	`--target c` (default)	Portable C99, float/transformer path plus exact moves.	Dynamic shapes it cannot bound.
non-FPU	`--target nonfpu`	Integer-only C, baked lattice tables, fixed-point math, no float/double/libm.	Any float path; a float_trap gate catches leaks.
CUDA	`--target cuda`	FP32 correctness-first kernels, one launch per op.	Missing toolkit or unsupported op.
Verilog	`jotnar-verilog`	Structural hardware for the supported integer subset.	Multi-lane arithmetic that would cross lanes; reassignment.
RV32IM	`jotnar-demo-doom`	A 32-bit machine model running a flat image.	Anything outside the RV32IM model.

The first three are selected by `jotnar build --target`. Verilog has its own binary because it needs a configuration file (widths, state, memories). The RV32IM path is a separate runner because it needs a compiled image and a WAD.

C

The C emitter is the portable baseline. It infers static shapes, emits array declarations, reads inputs from binary files in argv order, and writes outputs the same way.

```
jotnar emit-c showcase/language/saxpy.asm \
  --payload /tmp/saxpy.json --outputs OUT --out /tmp/saxpy.c
```

The emitted loop is exactly what you would write by hand:

```
/* ax = IMUL(A, X) 64-bit wrap */
for (int64_t e_i = 0; e_i < 4; ++e_i)
  ax[e_i] = (int64_t)((uint64_t)A[e_i] * (uint64_t)(X[e_i]));

/* OUT = IADD(ax, Y) 64-bit wrap */
for (int64_t e_i = 0; e_i < 4; ++e_i)
  OUT[e_i] = (int64_t)((uint64_t)ax[e_i] + (uint64_t)(Y[e_i]));
```

The casts are the wrapping contract. The `uint64_t` multiply is defined to wrap; the cast back to `int64_t` preserves the bit pattern. This is why the C target can claim exact integer parity across platforms.

STATIC SHAPES

The C emitter needs static shapes. If a dimension is dynamic, it returns a loud "dynamic dim" error rather than emitting a buffer it cannot bound. That is a feature: a silent fixed-size guess is how portable code becomes non-portable.

NON-FPU

The non-FPU target is the strict one. It emits C with no `float`, no `double`, and no libm. The lattice tables are baked into the source. Fixed-point arithmetic, matmul, norms, softmax, and the headt block are all integer-only.

```
jotnar build showcase/language/saxpy.asm \
  --payload /tmp/saxpy.json --outputs OUT --target nonfpu --workdir /tmp/nf
```

A source-level `float_trap` gate fails the build if a float slips into the generated code. That is the only way to keep the claim honest: the compiler cannot silently introduce a float if the source has none.

The non-FPU gates are exact for bigram decisions, T moves, fixed-point arithmetic, matmul, GT/SELECT, and the full headt block. That is a stronger claim than "close enough."

CUDA

The CUDA target is correctness-first. It emits one launch per operation, with FP32 kernels for the unit, bigram, and headt paths. It is not tuned; there are no CUDA graphs, no fused executor, and no cuBLAS scheduling.

```
jotnar build programs/lm_headt.asm \
  --payload payload.json --outputs LOGITS,OUT --target cuda --workdir /tmp/cu
```

The CUDA gates compare units within 1e-4 of numpy, bigram argmax bit-exactly, and the headt/fp16/D32/D64 paths against the reference agreement classes. `--fp16` reads F inputs from half-precision files while computing in FP32.

NOT A SPEED CLAIM

One launch per op is a correctness strategy, not a performance strategy. A tuned kernel pass is a separate project. If you benchmark this target and call it slow, you are right, and you have missed the point.

VERILOG

The Verilog backend emits structural hardware for the integer subset. It is not a CPU; it is the graph as combinational and sequential logic. The CPU8 fixture is the canonical example: a single-cycle 8-bit CPU expressed in Jotnar, emitted as Verilog, and checked against an independent ISA model.

```
jotnar-verilog tests/fixtures/cpu8.asm \
  --config showcase/verilog/cpu8.json --module cpu8
```

The config supplies widths, state registers, and memories. The backend refuses multi-lane arithmetic that would cross lanes and reassignment of a wire. Those are hardware constraints, not language constraints, and the emitter says so.

RV32IM

The DOOM runner is the largest integration test: a compiled RV32IM image executing on a Rust machine model with MMIO for WAD, framebuffer, clock, frame counter, key ring, and UART/CLINT. It is not an x86 emulator and not DOS.

```
./run_doom_demo.sh --frames 100
```

The frame-1 state is exact against the C native core; the frame-2 checkpoint is exact against the GPU core. That is a witness at the scale of a real program, not a toy loop.

AN EMITTER MAY CHANGE THE SCHEDULE. IT MAY NOT CHANGE THE ARITHMETIC. IF IT CANNOT PRESERVE THE CONTRACT, IT MUST REFUSE THE OP.

EMITTER RULE

14 PAYLOADS, CLI, AND THE WITNESS

The command line is the boundary between the graph and the world. It should be small enough to memorize and strict enough to trust.

The reference CLI has seven commands. Five of them are about a single listing; two are about the system.

COMMAND	WHAT IT DOES
`jotnar check <file.asm>`	Parse, expand, bind, and check arity.
`jotnar verify <file.asm>`	Replay layout unification; print coverage and the DEF interface.
`jotnar run <file.asm> --payload FILE`	Execute and write streams.
`jotnar emit-c <file.asm>`	Emit portable C99.
`jotnar build <file.asm> --target c|nonfpu|cuda`	Emit and build a native target.
`jotnar repeat <file.asm> --payload FILE`	Thread STATE streams across host iterations.
`jotnar ops`	Print the registry: name, arity, and whether a pattern exists.

Common flags: `--stdlib DIR` overrides the `IMPORT` search path; `--outputs a,b` selects streams; `--out FILE` writes to a file; `--json` makes `verify` machine-readable; `--fp16` switches CUDA inputs to half precision.

RUN

```
jotnar run showcase/language/saxpy.asm \
  --payload /tmp/saxpy.json --outputs OUT
```

The output is a JSON stream map. Without `--outputs`, every stream is written. Without `--out`, the JSON goes to stdout.

The payload is the only input surface. It has a `streams` object; each entry has a `kind`, a `shape`, and data. Here is the full vocabulary:

```
{
  "streams": {
    "f64": {"kind":"F64","shape":[2],"data":[0.5,-1.25]},
    "f32": {"kind":"F32","shape":[2],"data":[0.5,-1.25]},
    "i64": {"kind":"I64","shape":[2],"data":["9007199254740993","-4"]},
    "u8":  {"kind":"U8","shape":[3],"data":[0,127,255]},
    "b":   {"kind":"Bool","shape":[2],"data":[true,false]},
    "t":   {"kind":"T","shape":[2],
            "s":[1,-1],"e":[32000,33000],"z":[0,0]},
    "none":{"kind":"None","shape":[0],"data":[]}
  }
}
```

`F32` and `F64` are distinct. Integers may be numbers or strings; strings preserve int64 values above 2^53. `Bool` data must be `true` / `false`. `T` needs all three planes and they must match the shape.

THE PAYLOAD IS NOT A SUGGESTION

If a declared input is missing, the run fails and names the missing streams. If a stream's kind does not match the declared layout, the run fails. If a shape does not match an operation's rule, the run fails. The CLI does not fill defaults.

REPEAT

`jotnar repeat` is the host-driven loop. The payload is an iteration list:

```
{"iterations":[
  {"streams":{"x":{"kind":"I64","shape":[1],"data":["0"]}}},
  {"streams":{"x":{"kind":"I64","shape":[1],"data":["10"]}}}
]}
```

Non-`STATE` inputs are read from each iteration. `STATE` streams seed from iteration 0 and carry forward. `--grow a,b` allows append-only growth along axis 0. This is how a resident pipeline is driven from outside without giving the language a socket.

VERIFY

```
jotnar verify tests/fixtures/cpu8.asm --json
```

The JSON report includes the resolved layout map, the error list, the coverage fraction, and the DEF interface table. `verify` only replays declared layouts; it does not execute the program. A clean verify is a static claim, not a runtime witness.

THE TRUST SPINE

The parity harness is one kind of witness. The trust spine in Appendix A generalizes it: a certificate records the gate, the scope, the falsifiers, the nulls, and the bench labels. `jotnar-lab verify` is the command that checks the schema and the artifact hashes.

OPS

`jotnar ops` is the quickest way to answer "does this mnemonic exist?" It prints all 86 names, their arity, and a marker when a target has no reference pattern. If a name is missing, the assembler will reject it. There is no aliasing and no case-insensitive fallback.

THE PARITY HARNESS

The witness lives in `scripts/`. It needs the Python oracle checkout and `phi-core`:

```
export PHI_CORE_DIR=/path/to/phi-core
export JOTNAR_PY=/path/to/jotnar

python3 scripts/parity_ops.py      --bin target/release/jotnar
python3 scripts/parity_programs.py --bin target/release/jotnar
python3 scripts/parity_emit.py     --bin target/release/jotnar
python3 scripts/parity_cuda.py     --bin target/release/jotnar
```

`parity_ops.py` constructs a small listing per mnemonic, runs both implementations, and compares triples by plane, integers exactly, and floats with a tight tolerance. `parity_programs.py` runs the model programs. `parity_emit.py` compares emitted binaries. `parity_cuda.py` needs `nvcc` and a GPU.

THE COMMAND LINE IS NOT THE LANGUAGE. IT IS THE SMALLEST HONEST BOUNDARY BETWEEN THE GRAPH AND THE WORLD.

CLI RULE

END

THE DETERMINISTIC PRESS

IV

FIELD NOTES

Four listings, a CPU, a 43-op image pipeline on three targets, a DAC-to-RC-to-ADC analog field, DOOM, Futhark, protocol, and an honest analog filter.

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15 FOUR FIRST LISTINGS

The first fifteen minutes of Jotnar: a linear map, a reduction, a shared operand graph, and a branch-like select.

These four listings are the tour. They are small enough to run in a terminal and complete enough to show the language's shapes. The files live in `showcase/language/`.

RUN THE WHOLE TOUR

```
cargo run --release --bin jotnar-demo-tour
```

The tour runs all four and prints the results. You can also run each listing by hand with the CLI.

1. SAXPY

```
# SAXPY: A*X + Y.  A linear elementwise specimen.
IN A
IN X
IN Y

ax = IMUL(A, X)
OUT = IADD(ax, Y)
```

Payload:

```
{"streams":{
  "A":{"kind":"I64","shape":[4],"data":["3","3","3","3"]},
  "X":{"kind":"I64","shape":[4],"data":["1","2","3","4"]},
  "Y":{"kind":"I64","shape":[4],"data":["10","20","30","40"]}
}}
```

Expected: `[13, 26, 39, 52]`.

What to notice: `IMUL` and `IADD` are wrapping integer operations. The graph is two nodes deep. The output shape is the same as the input shape because both operations are elementwise.

2. REDUCE

```
# 8-way tree reduction.  A balanced fan-in-2 tree.
IN A0
IN A1
IN A2
IN A3
IN A4
IN A5
IN A6
IN A7

s0 = IADD(A0, A1)
s1 = IADD(A2, A3)
s2 = IADD(A4, A5)
s3 = IADD(A6, A7)
t0 = IADD(s0, s1)
t1 = IADD(s2, s3)
OUT = IADD(t0, t1)
```

Expected for inputs `0..7`: `28`.

What to notice: the tree is balanced. The graph has depth 3, not 7. The names `s0`, `s1`, `t0`, `t1` are descriptive, not special. A different naming would produce the same graph.

3. MATMUL2

```
# 2x2 matrix multiply. Four dot products with a shared operand graph.
IN a00
IN a01
IN a10
IN a11
IN b00
IN b01
IN b10
IN b11

t00 = IMUL(a00, b00)
t01 = IMUL(a01, b10)
c00 = IADD(t00, t01)
t10 = IMUL(a10, b00)
t11 = IMUL(a11, b10)
c10 = IADD(t10, t11)
t02 = IMUL(a00, b01)
t03 = IMUL(a01, b11)
c01 = IADD(t02, t03)
t12 = IMUL(a10, b01)
t13 = IMUL(a11, b11)
c11 = IADD(t12, t13)
```

For the identity times `[[1,2],[3,4]]`, expected `c00=1, c01=2, c10=3, c11=4`.

What to notice: the inputs are scalars, not matrices. The 2×2 structure is in the naming and the graph, not in a shape declaration. This is the smallest example of a shared operand graph: `a00` feeds both `t00` and `t02`.

4. SELECT

```
# A two-way select. One condition, two data inputs, no arithmetic.
IN C
IN A
IN B

OUT = ISELECT(C, A, B)
```

Expected: wherever `C` is nonzero, `OUT` takes `A`; elsewhere it takes `B`.

What to notice: `ISELECT` is not a branch. Both arms are values. The mask chooses elementwise. That is what makes the operation safe to reorder and easy to lower to hardware.

IF YOU CAN NAME THE NODES, YOU CAN READ THE GRAPH. IF YOU CAN READ THE GRAPH, YOU CAN RUN IT.

TOUR RULE

RUN ONE BY HAND

```
cat > /tmp/saxpy.json <<'JSON'
{"streams":{
  "A":{"kind":"I64","shape":[4],"data":["3","3","3","3"]},
  "X":{"kind":"I64","shape":[4],"data":["1","2","3","4"]},
  "Y":{"kind":"I64","shape":[4],"data":["10","20","30","40"]}
}}
JSON

cargo run --release --bin jotnar -- run showcase/language/saxpy.asm \
  --payload /tmp/saxpy.json --outputs OUT
```

The same listing can be lowered to C:

```
cargo run --release --bin jotnar -- build showcase/language/saxpy.asm \
  --payload /tmp/saxpy.json --outputs OUT --workdir /tmp/saxpy --name saxpy
```

END

16 A CPU IN ONE LISTING

If the language is worth its salt, you should be able to write a machine in it. Here is a single-cycle 8-bit CPU on a 16-bit datapath.

`tests/fixtures/cpu8.asm` is a complete single-cycle CPU. It has a 256×16 memory, two read ports (instruction and data), one clocked write port, a 16-bit accumulator kept 8-bit by masking, a zero flag, and a halt flag. It is written entirely in Jotnar integer operations. There is not a single branch in the listing.

Here is the header and the decode section:

CPU8 / TESTS/FIXTURES/CPU8.ASM

```
# cpu8: a single-cycle 8-bit CPU on the 16-bit Jotnar datapath.
# Word = [15:12] opcode, [7:0] operand.  Memory 256 x 16, two read ports
# (PC -> INSTR, ARG -> DATA) and one clocked write.  ACC is 16-bit but kept
# 8-bit by masking.  Z updates only on ALU ops, so STA preserves it.
#
#  0 HLT   1 LDA #   2 LDA addr   3 STA addr   4 ADD #   5 ADD addr
#  6 JMP   7 JZ      8 JNZ        9 INC       10 DEC    11 SUB #
#
# State: PC, ACC, Z, HALT (all 16-bit).  Reset clears them; memory keeps
# its init image and any writes.
IN PC
IN ACC
IN Z
IN HALT
IN INSTR
IN DATA

# fetch fields
OP = ISHR(INSTR, 12)
ARG = IAND(INSTR, 255)

# decode
H  = IEQ(OP, 0)
L1 = IEQ(OP, 1)
L2 = IEQ(OP, 2)
S3 = IEQ(OP, 3)
A4 = IEQ(OP, 4)
A5 = IEQ(OP, 5)
J6 = IEQ(OP, 6)
Z7 = IEQ(OP, 7)
N8 = IEQ(OP, 8)
I9 = IEQ(OP, 9)
D10 = IEQ(OP, 10)
B11 = IEQ(OP, 11)
```

The first two instructions split the 16-bit instruction word into a 4-bit opcode and an 8-bit operand. The twelve `IEQ` instructions turn the opcode into twelve one-hot signals. That is the decode stage.

The rest of the listing is a mux tree built from `ISELECT`:

```
# accumulator next (default: hold)
K0 = ISELECT(L1, ARG, ACC)
K1 = ISELECT(L2, D8, K0)
K2 = ISELECT(A4, C1, K1)
K3 = ISELECT(A5, C2, K2)
K4 = ISELECT(B11, C3, K3)
K5 = ISELECT(I9, C4, K4)
ACC_NEXT = ISELECT(D10, C5, K5)
```

Each candidate is computed unconditionally: `C1` is the add-immediate result, `C2` is the add-memory result, `C3` is the subtract result, and so on. The selects choose one. There is no branch, no speculative path, and no control hazard. The dataflow is the control.

THE REST OF THE MACHINE

The listing then computes:

- **The zero flag.** `Z_NEW = IEQ(ACC_NEXT, 0)`, and `Z_NEXT` updates only for ALU ops and loads. `STA` and `JMP` preserve it.
- **The halt flag.** `HALTED = IOR(HALT, H)`, and `HALT_NEXT` is its inverse-masked form.
- **The next PC.** `PC1 = IADD(PC, 1)`, `PC_TAKE = ISELECT(JT, ARG, PC1)`, and `PC_NEXT = ISELECT(HALTED, PC, PC_TAKE)`.
- **The write port.** `WE = IAND(S3, NH)`, `WDATA = IAND(ACC, 255)`, where `NH` is "not halted."

The memory is not in the listing. It is part of the Verilog configuration: depth 256, width 16, read address `PC`, read output `INSTR`, second read address `ARG`, second read output `DATA`, write enable `WE`, write address `ARG`, write data `WDATA`, and a 13-word init image. That is the machine boundary.

WHY THIS IS A GOOD TEST

The CPU exercises almost every integer mnemonic in combination: add, subtract, multiply? No — this CPU has no multiply. It exercises add, subtract, bitwise and/or/xor, shifts, compare, select, and masks. It exercises state. It exercises the difference between a value and a wire. If the lowering is wrong, the CPU does not boot.

EMIT THE HARDWARE

```
cargo run --release --bin jotnar-verilog -- \
  tests/fixtures/cpu8.asm \
  --config showcase/verilog/cpu8.json \
  --module cpu8
```

The backend emits structural Verilog and, with `cycles` and `trace` in the config, runs a simulation. The trace is checked against an independent ISA model. The config in the repo sets 90 cycles and a reset, which is enough to execute the tiny init image and show the fetch/decode/execute loop.

The same listing can be verified statically:

```
$ jotnar verify tests/fixtures/cpu8.asm
layouts resolved: 53/59 (0 error(s))
```

Fifty-three of fifty-nine streams have declared layouts. The rest are internal and inferred by the graph. Zero errors.

IF YOU CANNOT WRITE THE MACHINE AS A GRAPH, YOU CANNOT EMIT THE GRAPH AS A MACHINE.

HARDWARE RULE

END

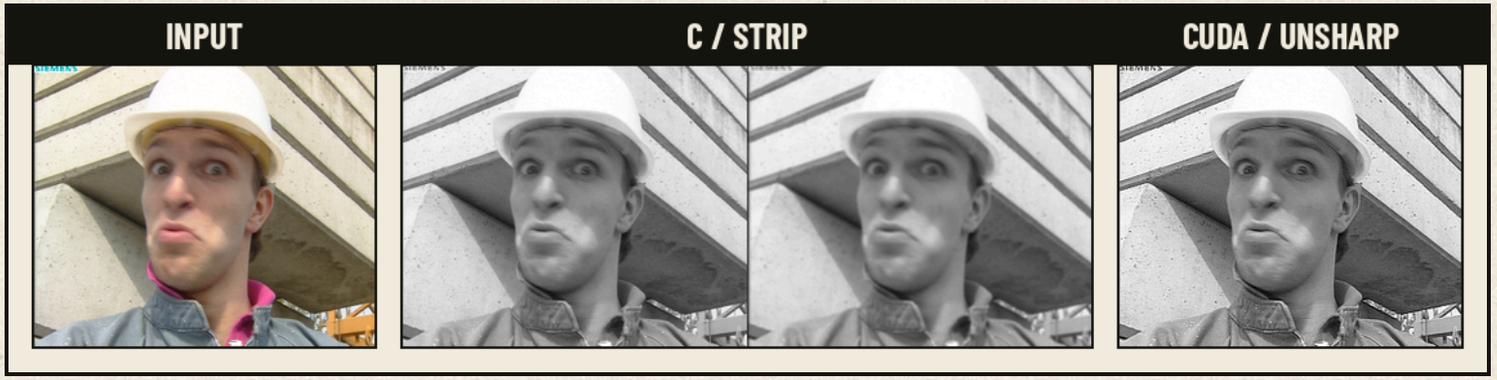
17 THE IMAGE PIPELINE

One 43-op listing, the same text on C, integer-only C, and CUDA. The image is the evidence, not the benchmark.

The image pipeline is the shortest way to see the same graph on three machines. It is one 43-op Jotnar listing, built by `src/showcase/image_demo.rs`, and it is byte-identical to the Python-generated listing. The C, non-FPU, and CUDA outputs are compared in the same run.

```
cargo run --release --bin jotnar-demo-img -- \
  --input samples/input_example.png --gallery gallery
```

The gallery contains the C output, the integer-only output, and the CUDA output. The source is a 352×288 photograph; the C strip shows the crop/blur path, and the CUDA image shows the unsharp result.



One listing, three targets: the input, the C crop/blur strip, and the CUDA unsharp result. The claim is the same graph, not pixel-identical floats on every platform.

WHAT THE RUN DOES

The listing decodes sRGB, computes luma, builds triples, applies the structure-tensor splat blur and beta field, applies a coherence-gated gain, and encodes back to sRGB. Chapter 10 explains the operations. This page is the exhibit claim.

TARGET	WHAT IT EMITS	WHAT THE RUN COMPARES
C	Portable C99, float/transformer path plus exact moves.	The C gallery image and the Python emitter output.

non-FPU	Integer-only C with baked lattice tables; no float, no double, no libm.	The integer gallery image and the source-level float trap.
CUDA	FP32 correctness-first kernels, one launch per op.	Units within 1e-4 of numpy; decisions and exact moves.

The three outputs are not required to be bit-identical everywhere. Float `F` paths can differ from numpy in the last ulp on some platforms because of a different SIMD reduction order. Triples and integer paths are compared bit-exactly; exact moves are exact.

> **WHAT THE FIGURE IS NOT**
>
> The figure is a visual check, not a witness. The witness is the comparison in `scripts/parity_emit.py` and the gallery run. A pretty image can hide a wrong stream; a byte comparison cannot.

WHAT IT PROVES

- The same 43-op listing can be lowered to three different targets without changing the graph.
- The exact moves and integer/triple paths survive the lowering.
- The non-FPU target can run the image path without a float in the generated source.
- The CUDA target is correct-first: one launch per operation, no tuned-kernel claims.

WHAT IT DOES NOT PROVE

- **Hardware speed.** The CUDA backend is correctness-first. A correctness run is not a performance result.
- **Every opcode on every backend.** The image listing is a selected 43-op path, not the full 86-mnemonic register.
- **Pixel identity across every platform.** Floats can differ in the last ulp; the comparison classes are explicit.
- **That the image is what you meant.** The gate checks the lowering against the reference, not the artistic intent.

THE IMAGE IS THE EVIDENCE THAT THE GRAPH SURVIVED. IT IS NOT THE PROOF THAT THE GRAPH IS RIGHT.

IMAGE EXHIBIT RULE

END

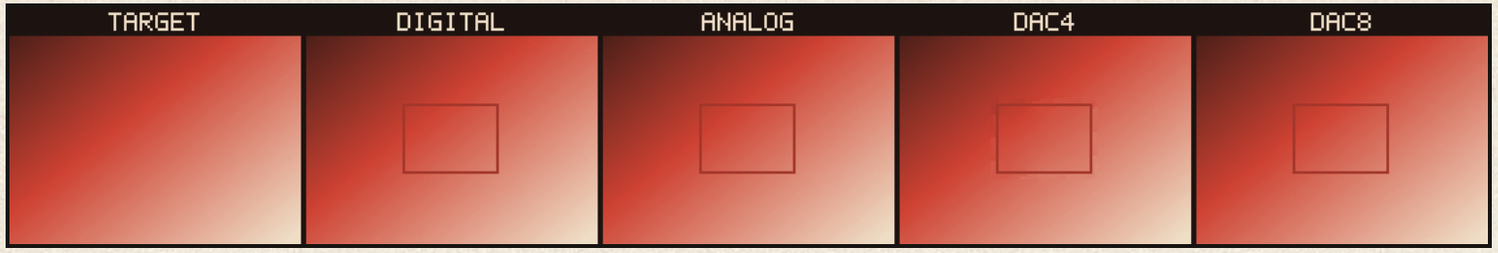
18 THE ANALOG FIELD

DAC → 2D RC field → ADC. A boundary-value operator, not a metaphor.

This is the **analog field** exhibit, not the analog filter of the next chapter. It is the smallest honest version of the analog story: encode a boundary with a DAC, let a 2D RC field settle, read it back with an ADC, and gate the result against digital Laplace.

```
cargo run --release --bin jotnar-demo-analog -- --out showcase/analog/out
```

The run writes `panel.png`, `metrics.json`, and `rjm_field2d.research.json`. The checked-in panel is a reference run.



TARGET | DIGITAL | ANALOG | DAC4 | DAC8. The digital and ideal-RC fields settle to the same harmonic solution; DAC resolution appears as quantization error. The duotone is for the zine; the numbers are in the table.

WHAT THE PANEL SHOWS

1. the target linear field;
2. a digital Laplace/Jacobi fill;
3. the ideal RC field;
4. RC with a 4-bit PCM DAC on the boundary;
5. RC with an 8-bit PCM DAC.

The digital and ideal-RC fields settle to the same harmonic solution. DAC resolution is directly visible as quantization error. That is the point: the field is a real boundary-value operator, not a metaphor.

THE NUMBERS

On the reference 80×112×3 grid, 999 hole pixels:

METHOD	HOLE L1
Digital Laplace	1.39e-6
Ideal RC	1.39e-6
RC + 2-bit DAC	0.0491
RC + 4-bit DAC	0.00553
RC + 6-bit DAC	0.000466
RC + 8-bit DAC	0.000117

At 8-bit DAC, ADC resolution and noise are also observable. The 8/10-bit operating point passes the tolerance gate: `0.000266 <= 0.001`.

> **THE GATE IS TOLERANCE, NOT BIT-EXACT**
>
> The analog field gate is `"class": "tolerance"`, `"bit_exact": false`, `"provenance": "simulated"`. It is not a bit-exact Jotnar-MAP contract. The field is a simulated analog substrate; the gate says so.

LIVE PATH VS. BAKED MAP

The live path needs a DAC because it is programming the boundary of a simulated field. A **live no-DAC** field is correctly refused:

```
"live_no_dac": {"accepted": false, "missing": ["dac.pcm"]}
```

The fast path is different. Solve once, store the settled field as a **Refined Jotnar Map**-shaped artifact with provenance, then apply it as a **Jotnar-VEC**-shaped lane batch over K patches. The artifact carries both profiles:

```
"live":      {"accepted": true, "mode": "dac_rc_adc"}
"baked_map": {"accepted": true, "mode": "apply_refined_map"}
```

That is storage and compute separated: the map is data at rest; the vector is work in motion. It is not a license to call the live solve DAC-free.

WHAT IT PROVES

- A 2D RC field reaches the digital Laplace solution in the ideal limit.
- DAC/ADC bits, settling, and noise are observable and sweepable.

- A capability gate can admit a field map and refuse a live path whose required primitive is absent.
- One map can be applied as a lane vector across many patches.

WHAT IT DOES NOT PROVE

- **Physical analog speed, energy, or device accuracy.** This is Jotnar-SIM, not Jotnar-ANA.
- **That RC replaces surface completion.** It does not.
- **That the bit-exact RJM rule accepts this artifact.** The gate is explicitly a tolerance gate.
- **That a simulated wall-clock is hardware time.** It is not.

The honest hybrid target remains **digital geometry + analog boundary conditioning + digital readout/gate.** The field supplies low-frequency continuity; it does not invent the missing geometry.

THE FIELD IS A BOUNDARY-VALUE OPERATOR. IT IS NOT A COMPLETION MODEL, AND A SIMULATED SECOND IS NOT A HARDWARE SECOND.

ANALOG FIELD RULE

END

19 FIELD NOTES: FOUR EXHIBITS

Four exhibits that are deliberately different: a real game, a word-level inference unit, a model-blind fleet, and an honest analog filter.

A language is only as convincing as the things people build with it. Part IV is a **selected set**: the 43-op image pipeline and the DAC → RC → ADC analog field have their own chapters (17 and 18). These four exhibits are not benchmarks. They are stories about the same stack under different pressures.

DOOM ON THE RV32IM CORE

A real compiled program on a real machine model. `doomgeneric` is built with Zig clang/lld for `riscv32-freestanding -mcpu=generic_rv32+m`, producing an 847,112-byte flat image. The Rust core implements RV32I + M, the CSR/trap surface, and the MMIO the game needs: WAD, framebuffer, clock, frame counter, key ring, UART/CLINT.

```
frame 1: 24,502,445 instructions, pc=0x00048860
framebuffer: BYTE-IDENTICAL
```

Frame 1 is exact against the C native core. The frame-1 checkpoint can be resumed by the C core, and the frame-2 state is byte-identical to the GPU core's checkpoint and framebuffer. The player is playable; it is not yet smooth. That is an honest known issue, not a correctness issue.



The title screen, halftoned. The image is not the witness; the framebuffer bytes are.

FUTHARK: ONE WORD, ONE LISTING

The Futhark exhibit treats a word as a request to a Jotnar listing. The host encodes characters as integer IDs and renders rune codepoints; the translation decision lives in the program.

The listing contains three things:

1. a 128-bit fingerprint index (FNV-1a plus a polynomial) over a fixed 28-character word, searched in 17–20 binary-search steps;
2. the JavaScript greedy spelling rules compiled into a dense action table over a four-character window, remaining length, `has_output`, and `position==0`;
3. a rune table mapping each phoneme to up to eight codepoints.

Correctness is gated at **118/118 words**, including trained words, exception contractions, and unknown words that exercise the fallback table. The Python oracle was itself checked 118/118 against the actual JavaScript in headless Chromium before it was used as the oracle.

The boundary is explicit: DOM traversal, text splitting, settings, fonts, and the final separator stay host-side. Jotnar does the decision, not the browser.

PROTOCOL: SAME BYTES, ANY ORDER

The protocol exhibit keeps the contract and drops the network transport. Nodes are real worker threads behind the same frame codec: `u32BE || JSON header || payload`.

A node never sees “Futhark,” “matmul,” or a model name. It receives a listing, named blobs, an `EXEC` binding of input stream names to blobs or inline byte segments, and output names to return.

The controller runs the same 12 jobs in original order, reverse order, and deterministic shuffled orders, with different node placements and simulated jitter. The completion order changes. The per-job output bytes do not:

```
original: completion=[0, 11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1] exact=12/12
reverse:  completion=[11, 0, 3, 6, 9, 1, 4, 7, 10, 2, 5, 8] exact=12/12
shuffle:  completion=[4, 0, 3, 7, 5, 6, 11, 10, 1, 9, 8, 2] exact=12/12
determinism: 3 batches, 12 jobs, 4 node assignments, byte-identical
RESULT: ALL OK
```

The interesting contract is not TCP. It is: **same program + same input bytes = same output bytes, independent of placement, submission order, and completion order.**

ANALOG: THE HONEST FILTER

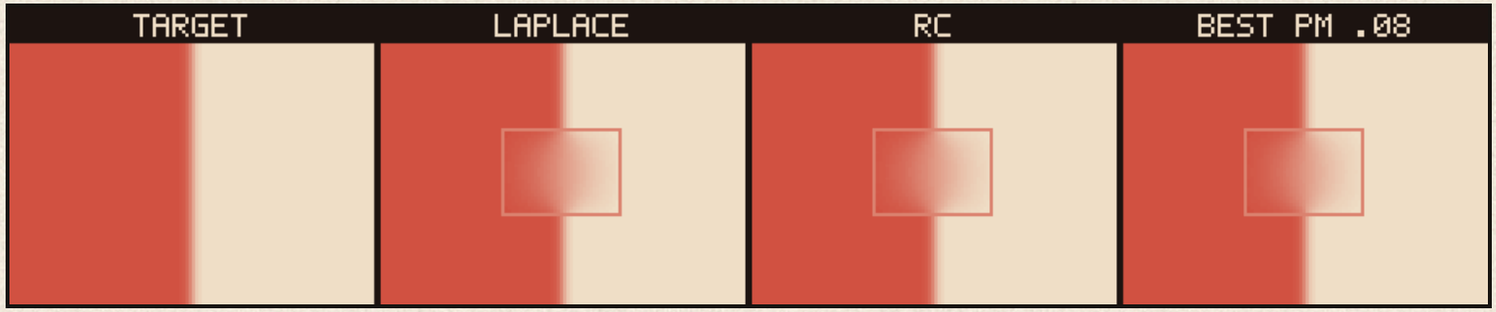
The analog-filter exhibit asks whether a Perona-Malik anisotropic diffusion can help an image problem better than digital Laplace. It is a simulated PDE experiment, **not** the DAC → RC → ADC field of Chapter 18. It is not Jotnar-ANA, and it makes no claim about physical hardware speed.

FILTER	HOLE L1	EDGE-BAND L1	SIMULATED TIME
Laplace	0.0715	0.1181	~0.034 s
RC	0.0715	0.1181	~0.266 s
PM exp k=0.08	0.2380	0.1801	~0.478 s
PM rational k=0.08	**0.0686**	**0.1128**	~0.344 s
PM rational k=0.03	0.2072	0.2011	~0.343 s
PM rational k=0.16	0.0708	0.1171	~0.342 s

The best synthetic gain is about 4% relative on the hole and 4.5% on the edge band, at roughly **10×** the Laplace solve time. On the real self-occlusion hole, PM is **worse** (0.1777 vs. 0.1729). A local PDE cannot invent the missing sphere side.

BENCH LABELS

These are a representative current run on this host. Bench labels: `clock=host_wall` (measured), `scope=end_to_end`, `kind=measured`. The field is simulated, not measured analog hardware. `showcase/analog_filter/metrics.json` is the authority if the numbers move again.



TARGET | LAPLACE | RC | BEST PM .08. The red outline is the hole boundary. The duotone is for the zine; the numbers are in the table.

The useful interpretation is the same as the earlier work: not the fill itself; possibly boundary/seam correction around a surface completion; possibly edge-aware depth/JBU filtering; possibly temporal integration. The positive hybrid remains digital surface geometry plus Laplace seam correction.

WHY INCLUDE A NEGATIVE RESULT?

Because a manual that only shows wins is marketing. The analog filter is a real experiment with a real limit. The stack is more credible when it reports the limit in the same voice as the win.

A DEMO IS A CLAIM WITH A WITNESS. IF THE WITNESS IS MISSING, IT IS A POSTER.

SHOWCASE RULE

END

V

APPENDICES

The trust spine, the press, the vocabulary, and the end matter.

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A THE TRUST SPINE

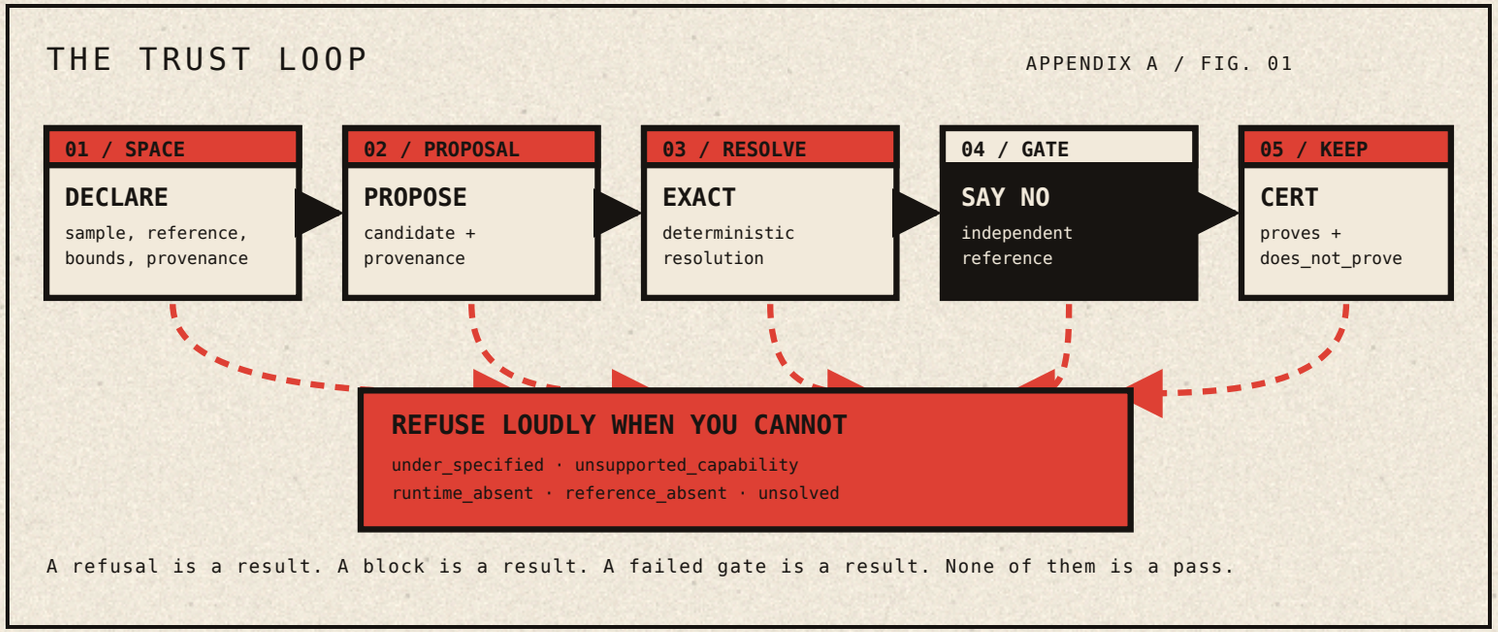
A demo is not a claim. The trust spine records what a result proves, what it does not, and when to refuse.

The trust spine is **not a language feature**. It is the development harness that lets us say a claim about a program, a machine, a map, a vector, a plan, or a certificate is true. Jotnar-LAN defines programs; the trust spine defines the discipline for claiming something about them.

The rule is one sentence:

> Construct the small space, propose, resolve exactly, gate independently, keep the certificate — and when you cannot, refuse.

The unity is in the loop, not in the artifacts. A demo is an instance of the loop at one layer. The normative summary is `docs/LOOP.md`; the schema is `docs/appendix/TRUST_SPINE.md`; the implementation is `crates/jotnar-lab`. This appendix is the reader's version.



The trust loop. Every stage may stop; a refusal, a block, and a failed gate are results, but none of them is a pass.

THE LOOP

```
construct a small space
        |
        v
propose a candidate
        |
        v
resolve it exactly
        |
        v
gate it independently
        |
        v
keep the certificate
        |
        +--> refuse loudly when the space, data, or runtime is absent
```

TERM	MEANING
Space	The declared sample space, reference or bound, and provenance. It is part of the contract.
Proposal	The candidate produced by a proposer, with provenance.
Gate	An independent predicate on (map, sample, reference). It must be able to say no.
Witness	A fact: map, input, output, trace.
Proof	A witness plus an independent gate plus a declared contract.
Derivation	The trace that makes the answer legible.
Certificate	The recorded witness/proof/derivation, with scope, unknowns, falsifiers, nulls, provenance.
Refusal	A typed statement that the process could not produce a proposal.

A gate is **not** `m == P(spec)`. That is vacuous: it restates the proposal. A real gate has its own reference and can reject a wrong proposal. The certificate records `gate.can_say_no`; a gate that cannot say no is not a gate.

THREE AXES, NOT ONE

AXIS	VALUES	WHAT IT MEANS
Verdict	pass, fail, bottom	The gate’s decision. bottom means the gate could not decide. It is never a pass.

Outcome	solved, refused, blocked, not_tested	The process result. A correct refusal is a successful certificate. A block is an honest certificate. A skipped control is not_tested.
Status	current, stale, historical, blocked, superseded	The certificate's standing. A stale or historical certificate is evidence about the past, not a current claim.

A valid certificate can therefore be:

```
outcome = refused
verdict = bottom
status  = current
```

`verify` must accept that as a successful refusal — while still refusing to call it a pass.

THE FIRST RULE OF THE SPINE

A blocked run is not a pass. A skipped control is not a pass. A gate that cannot say no is not a gate. A certificate without `does_not_prove` is an overclaim.

GATE INDEPENDENCE AND THE VACUITY TEST

The gate must be able to reject the proposal for the right reason. The vacuity test is the cheapest way to catch a fake gate:

```
replace the gate with   m == P(spec)
show that a wrong map still passes
```

If the wrong map **does not** pass, the test is broken. The DOOM certificate records this as a falsifier row: "replace gate with `m == P(spec)`" → wrong map passes → `not_caught` by construction. That is the independence test, not a failure of the gate.

REQUIRED HONESTY FIELDS

`Certificate::validate` rejects a certificate that omits:

- `proves` or `does_not_prove` (an empty boundary is an overclaim);
- declared assumptions and unknowns;
- the gate hash and `can_say_no = true`;
- at least one falsifier row and one null row (`not_tested` is allowed);
- bench `clock`, `scope`, and `kind` labels;
- provenance (command, cwd, commit, host, environment) and artifact paths/hashes.

A certificate with `outcome = refused` must carry a refusal. A certificate with `status = superseded` must carry `superseded_by`. `certificate_sha256` is computed over canonical JSON without the hash field, then written; `verify` recomputes it.

FALSIFY, NULL, BENCH

They are duals, not one bucket.

TOOL	TESTS	HONEST NON-RESULT
Falsify	The gate's sensitivity: corrupt the proposal and see whether the gate catches it.	not_caught names the trust boundary.
Null	The claim's specificity: run a control that should not produce the effect.	not_tested is an honest state; a confounded control is a finding.
Bench	The cost claim: frozen inputs, repeated runs, a distribution, a worst case.	A benchmark is a cost certificate, not a leaderboard.

Bench labels are mandatory:

```
clock: rdcycle (emulated) | host_wall (measured) |
       cuda_event (measured) | analog (simulated) |
       calibrated (0.185 ns/node)
scope: apply | total | startup | end_to_end | worst_frame
kind:  measured | emulated | simulated | calibrated
```

A simulated analog wall-clock is not measured hardware time. The label says which one it is.

THE DOOM CERTIFICATE

The real certificate is `certificates/doom/native-gpu-migration.json`. It records:

- frame 1: Rust vs. C native on the first 64 MB, registers, PC, cycles, and framebuffer;
- frame 2: Rust checkpoint vs. the GPU checkpoint on the first 64 MB, registers, PC, and framebuffer;
- **does not prove**: the GPU kernel for all programs, plan completeness, memory beyond 64 MB, or fleet throughput.

The falsifier table deliberately includes boundary rows:

CORRUPTION	EXPECTED	ACTUAL	NOT_CAUGHT
framebuffer byte changed	fail	fail	false
register byte changed	fail	fail	false
PC changed	fail	fail	false

memory byte changed (no frame effect)	fail	fail	false
memory byte changed beyond 64 MB	fail	pass	**true**
compare 128 MB native with 64 MB GPU	fail (space mismatch)	fail (full) / pass (64 MB slice)	**true**
wrong frame	fail	pass (frames identical)	**true**
replace gate with m == P(spec)	pass (wrong map must pass the vacuous gate)	pass	**true**

The null controls include `wrong_frame` (expected reject, actual pass because the title frame is static; status `not_tested`) and `shifted_frame` (expected reject, actual reject; status `pass`). The bench is `clock=host_wall`, `scope=end_to_end`, `kind=measured`, three runs, with p50/p99/max.

THE DOOM CERTIFICATE / CONDENSED

```
schema:          jotnar.certificate
status:          current
outcome:         solved
verdict:         pass
gate:            byte_identical_migration
can_say_no:      true
proves:          Rust frame 1 matches C native on 64 MB/regs/PC/cyc/fb;
                 Rust frame 2 matches the GPU checkpoint on 64 MB/regs/PC/fb
does_not_prove:  GPU kernel for all programs; plan completeness;
                 memory beyond 64 MB; fleet throughput
falsifiers:      8 rows, including 4 not_caught (space boundaries + vacuity)
nulls:           2 rows, including wrong_frame = not_tested (static title)
bench:           clock=host_wall, scope=end_to_end, kind=measured
```

VERIFY, AND WHY BLOCKED IS NOT PASS

`verify` checks the schema, the certificate hash, and the artifact hashes by default. `--no-hashes` skips only the artifact check; the schema and certificate hash are still checked.

CERTIFICATE STATE	VERIFY SAYS
outcome=solved, verdict=pass, status=current, artifacts present	PASS
outcome=refused with a refusal	REFUSED (valid)
outcome=blocked	BLOCKED
outcome=not_tested	NOT TESTED
verdict=fail	FAIL (certified negative)
verdict=bottom	BOTTOM (gate could not decide)

status=stale/historical/superseded	STALE / HISTORICAL / SUPERSEDED (not a current claim)
artifacts missing	BLOCKED, exit non-zero

The self-contained example needs no external assets. Run these commands from the Jotnar repository root (the parent directory):

```
cargo run -p jotnar-lab -- verify certificates/example/certificate.json
```

The DOOM certificate references local `/tmp` artifacts. On this host it verifies **PASS**. Without those artifacts it prints **BLOCKED** and exits non-zero. That is intentional: a certificate about files that are no longer present is not a pass. A caller may choose to accept a block for its own purpose, but that is the caller's policy, not the certificate's claim.

The real commands, also from the Jotnar repo root:

```
# Build the bundle from the native/C/GPU parity run.
python3 scripts/parity_doom_native.py \
  --gpu-work /tmp/opencode/doom_overlay \
  --bundle /tmp/opencode/doom_bundle

# Read existing artifacts; do not re-run the CPU.
cargo run -p jotnar-lab -- certify doom \
  --bundle /tmp/opencode/doom_bundle/manifest.json \
  --out certificates/doom/native-gpu-migration.json

cargo run -p jotnar-lab -- falsify doom \
  --bundle /tmp/opencode/doom_bundle/manifest.json

cargo run -p jotnar-lab -- null doom \
  --bundle /tmp/opencode/doom_bundle/manifest.json

cargo run -p jotnar-lab -- bench doom \
  --bundle /tmp/opencode/doom_bundle/manifest.json \
  --runner target/release/jotnar-demo-doom --runs 3 \
  --certificate certificates/doom/native-gpu-migration.json
```

STATUS AT THIS DRAFT / 2026-11-06

Workspace tests: **61/61**. Self-contained example certificate: **PASS**. DOOM certificate: **PASS on this host**, and **BLOCKED** when the local artifacts are absent. Manual witness: **4/4** listings. The trust spine does not make every Jotnar claim certified; it makes the claim's boundary explicit.

WHAT THE HARNESS MUST NOT DO

- It must not call a blocked run a pass.

- It must not accept a certificate without `does_not_prove`.
- It must not accept a gate that cannot say no.
- It must not turn a vacuous gate into evidence.
- It must not require a positive verdict to call a certificate valid; an honest fail, refusal, or block is a certificate too.

A CERTIFICATE IS A CLAIM WITH ITS BOUNDARY ATTACHED. IF THE BOUNDARY IS MISSING, THE CLAIM IS NOT CERTIFIED.

TRUST SPINE RULE

END

B BUILDING THIS MANUAL

The book is a build artifact. Here is the pipeline, the source, and the reason it is Rust rather than Jotnar.

This manual is built from Markdown files by a small Rust program called **jotnar-press**. The press reads `book.toml`, renders each chapter, generates the cover, masthead, table of contents, part dividers, and the 86-mnemonic register, then hands the HTML to Paged.js and Chromium for pagination and PDF output.

```
content/*.md
   -> jotnar-press (pulldown-cmark + directives)
   -> out/manual.html
   -> Paged.js (page boxes, running heads, page numbers)
   -> Chromium (print-to-PDF, CSS @page)
   -> out/jotnar-field-manual.pdf
```

The whole thing is deliberately inspectable. There is no layout engine to learn, no binary template, and no proprietary typesetting step. If a page is wrong, it is wrong in the Markdown, the CSS, or the builder.

THE TREE

PATH	WHAT IT IS
book.toml	Book metadata, part order, chapter order, themes.
content/*.md	One Markdown file per chapter, with a small frontmatter block.
data/ops.tsv	The 86-mnemonic register: name, arity, signature, family, one-line contract.
press/src/main.rs	The press: frontmatter, directives, syntax highlighting, HTML, PDF.
press/vendor/paged.polyfill.js	Paged.js 0.4.3, vendored so the build is offline.
theme/book.css	The zine theme: page boxes, type, color, code, tables, cover.
assets/	Fonts, textures, halftone images, diagrams, cover art.
scripts/make_assets.py	Deterministic texture and halftone generator.
scripts/witness.sh	Runs the four first listings through the Jotnar interpreter.
out/	Generated HTML, PDF, and preview images.

BUILD

```
cargo run --manifest-path press/Cargo.toml --release -- \
  --preview
```

The `--preview` flag writes 72-dpi PNGs to `out/preview/` after the PDF. `--html-only` stops before Chromium; `--no-pdf` is the same; `--out` changes the PDF path.

ADD A CHAPTER

1. Create `content/22-my-chapter.md` with frontmatter:

```
---
title: My Chapter
number: "18"
kicker: FIELD NOTE / EXPERIMENT
lede: One sentence that tells the reader why this chapter exists.
columns: 2
---
```

2. Add the file to the appropriate `[[part]]` in `book.toml`.
3. Rebuild. The TOC page number is filled by the Paged.js `after` hook, so you do not need to edit it.

The frontmatter keys are: `title`, `number`, `kicker`, `lede`, `columns` (1 or 2), `feature` (true for a full-page opener), `theme` (`red`, `dark`, or `paper`), `art`, `art_caption`, and `stamp`.

DIRECTIVES

The press adds a few blocks on top of Markdown. They are deliberately plain:

DIRECTIVE	RESULT
`::: note [TITLE]`	A cream note box with a red rule.
`::: warning [TITLE]`	A black warning box with a red rule.
`::: sidebar [TITLE]`	A red sidebar box.
`::: pullquote [ATTRIBUTION]`	A large uppercase quote with a citation.
`::: figure src="..." caption="..."`	A figure with a caption; class="bleed" for full width.
`::: listing [CAPTION]`	A code block with a label.
`::: op-table family="int"`	A generated table for one family, or all.

`::: stamp [TEXT]`	A rotated stencil stamp.
`::: end`	The black END mark.
`::: pagebreak`	A forced page break.

Markdown inside a directive is rendered normally, so a note can contain lists, code, and tables.

THE BUILDER, IN ONE FUNCTION

The heart of the press is a small block renderer. It receives the directive name, the attributes, and the rendered body:

```
match name {
    "note" => format!(
        r#"<aside class="note span-all"><div class="aside-title">{}</div>{inner}</aside>"#,
        escape(&label)
    ),
    "warning" => format!(
        r#"<aside class="warning span-all"><div class="aside-title">{}</div>{inner}</aside>"#,
        escape(&label)
    ),
    "listing" => format!(
        r#"<div class="listing span-all"><div class="listing-label">{}</div>{inner}</div>"#,
        escape(&label)
    ),
    _ => format!(r#"<div class="block {name} span-all">{inner}</div>"#),
}
```

The `span-all` class is kept for the two-column experiments; the current prose grid is a measured single column. The rest is CSS.

THE PAGINATION HOOK

Paged.js handles the page boxes, the running heads, and the page counter. The only dynamic piece is the TOC page number. The press injects a tiny `after` hook:

```
window.PagedConfig = {
  auto: true,
  after: function () {
    var pages = [].slice.call(document.querySelectorAll('.pagedjs_page'));
    var pageOf = new Map();
    pages.forEach(function (page, idx) {
      page.querySelectorAll('[id]').forEach(function (el) {
        if (!pageOf.has(el.id)) pageOf.set(el.id, idx + 1);
      });
    });
    document.querySelectorAll('.toc-page[data-target]').forEach(function (el) {
      var n = pageOf.get(el.getAttribute('data-target'));
      if (n) el.textContent = String(n).padStart(2, '0');
    });
  }
};
```

That is the entire coupling between the content and the pagination. There is no two-pass build and no manual page-number table.

WHY NOT WRITE THE PRESS IN JOTNAR?

This is the honest answer: Jotnar is a dataflow language over typed streams. Its boundary is a payload of streams and a set of named outputs. It has no filesystem, no string type, no Markdown parser, and no subprocess. Writing a Markdown-to-PDF builder in Jotnar would mean building an operating system, a text model, and a font rasterizer first. That is a different project, and a good one, but it is not this one.

What Jotnar *can* do is witness the examples. `scripts/witness.sh` runs the four first listings through the reference interpreter and checks the outputs the manual quotes. The press builds the book; Jotnar checks the claims the book makes about itself.

> **THE HONEST SPLIT**
>
> Host code does I/O, parsing, and rendering. Jotnar does the typed computation. The boundary is the payload. Keeping that split explicit is the same discipline the language asks of its users.

MAKE IT YOURS

The theme is a single CSS file. The tokens at the top control the paper, ink, red, and the font stacks. The cover is generated from `book.toml` plus `assets/cover-art.svg`. The part dividers use `theme = "red" | "dark" | "paper"` in `book.toml`. The halftones and textures are regenerated deterministically by `scripts/make_assets.py`.

If you change the page size, change `@page { size: ... }` and the fixed-size cover/part sections together. If you change the font, add a `@font-face` and update the token. If you change the register, edit `data/ops.tsv`; the chapter body is only a directive.

“

A MANUAL YOU CAN REBUILD IS A MANUAL YOU CAN TRUST.

PRESS RULE

END

C GLOSSARY

The words are load-bearing. Use them the way the stack uses them.

TERM	MEANING
arity	The number of inputs and outputs a mnemonic requires. Wrong arity is an assembly error.
assembler	The frontend that expands, parses, binds, and verifies a listing.
bench	A cost certificate: frozen inputs, repeated runs, a distribution, and a worst case, with mandatory clock, scope, and kind labels. Not a leaderboard.
blocked	An outcome or status: the artifacts, runtime, or reference are absent. Honest, never a pass.
bottom	A gate verdict: the gate could not decide. Never a pass.
bound program	A parsed program with each instruction bound to a registry signature.
CALL	Textual expansion of a DEF at a call site, with namespaced internal names.
certificate	The recorded witness/proof/derivation, with scope, unknowns, falsifiers, nulls, bench labels, and provenance.
coherence	A local orientation measure in [0,1] produced by the structure tensor.
CONFIG	A program-global constant such as m_acc, m_cov, or beta.
DEF	A named reusable subgraph with formals and declared outputs.
derivation	The trace that makes an answer legible.
DIG	Digital Jotnar: exact, bit-identical, C/CUDA/RV64.
does_not_prove	The required boundary of a certificate's claim. An empty boundary is an overclaim.
emitter	A backend that lowers a bound program to a target artifact.
fail loud	The doctrine that errors are errors, never silent guesses.
falsifier	A test of the gate's sensitivity: corrupt the proposal and see whether the gate catches it.
field	A spatial stream/lattice: depth, normals, IDs, provenance. Jotnar-FLD.
fixed-point bridge	to_fixed / from_fixed: the conversion between triples and int64 at a scale.
gate	An independent predicate on (map, sample, reference). It must be able to say no.

gate independence	A gate with its own reference, not m == P(spec). A gate that cannot reject a wrong proposal is not a gate.
Jotnar	The family/stack: program, machine, witness. Not a project name.
Jotnar-LAN	The language/IR/op set/plan. The program.
Jotnar-MAC	The machine/target: ISA, memory, lanes, OS, board.
Jotnar-WIT	The witness/contract: trace, certificate, provenance, byte-identity.
kind	F32, F64, I64, U8, Bool, Triple, None.
layout	A KIND:GEOM string such as T:HW or F:HWC.
lattice	The phi-value substrate: sign, exponent, zero flag; integer arithmetic in the middle.
map	A stored field/artifact: shape, dtype, layout, hash. Jotnar-MAP.
mnemonic	An uppercase instruction name, bound to the registry.
NoPattern	A loud emitter error: the target has no lowering for this mnemonic.
not_caught	A falsifier row the gate did not catch. It names the trust boundary; it is not dishonesty.
not_tested	A deliberately skipped control or outcome. Honest, not a pass.
null control	A deliberate control that should not produce the effect. It tests the claim's specificity.
outcome	solved, refused, blocked, or not_tested. The process result, separate from the gate verdict.
payload	The JSON boundary: a streams map of typed input values.
plan	The executable binding of maps (inputs/outputs) to vectors (work). Jotnar-PLN.
proof	A witness plus an independent gate plus a declared contract.
proposal	The candidate produced by a proposer, with provenance.
refusal	A typed statement that the process could not produce a proposal. A refusal is a result.
registry	The authoritative table of 86 mnemonics, arities, and signatures.
shape	The dimensions of a stream, e.g. [4], [H,W], [H,W,C].
signature	A layout pattern per input and output, with $VAR, *, and literals.
space	The declared sample space, reference or bound, and provenance. Part of the contract.
STATE	A stream carried across host iterations by jotnar repeat.
status	current, stale, historical, blocked, or superseded. A stale or historical certificate is not a current claim.
stream	A name for a value with a kind and a shape. The central object.
triple	Three planes: sign, exponent, zero flag. A scale-free lattice value.
unification	Matching a signature pattern against an actual layout, binding $VARs.

vacuity test	Replace the gate with m == P(spec) and show that a wrong map still passes. If it does not, the test is broken.
vector	A costed batch of data-parallel work. Jotnar-VEC.
verdict	pass, fail, or bottom. The gate's decision, separate from the outcome.
witness	A fact: map, input, output, trace. A witness is not a proof until an independent gate and a contract are attached.

NAMING DISCIPLINE

When a sentence uses "Jotnar" without a qualifier, it means the family/stack. A specific claim names the aspect: **Jotnar-LAN**, **Jotnar-MAC**, or **Jotnar-WIT**. A digital/analog claim names the flavor: **DIG**, **SIM**, **ANA**, or **HYB**.

END

D COLOPHON

Who made this, what it is set in, and what it is licensed under.

JOTNAR FIELD MANUAL — Issue 01, Digital, First Rough Draft. November 2026.

Published by **The Deterministic Press** for the Jotnar Working Group.

EDITORIAL

This is a first rough draft of a manual for a language under active development. The examples are real, the gaps are real, and the assembler is the authority. Corrections should be made against the source, not the PDF.

TYPE AND TOOLS

ELEMENT	CHOICE
Display	Anton
Condensed labels	Barlow Condensed
Body	Courier Prime
Code	IBM Plex Mono
Masthead / quotes	Special Elite
Stamps	Stardos Stencil
Numbers	Bebas Neue
Setting	Markdown, Rust, Paged.js 0.4.3, Chromium
Textures and halftones	Deterministic PIL generator (scripts/make_assets.py)
Page size	840 × 660 pt (11.67 × 9.17 in), landscape

The fonts are open-licensed and vendored in `assets/fonts/`. The page size is deliberately the same landscape format as a folded zine spread. The cover, part dividers, and feature openers are full-bleed; the content pages use a measured single column with a red rail, and the contents/register pages use their own grids.

LICENSES

The Jotnar implementation and the code excerpts in this manual are **GPL-3.0-only**, matching the upstream project. The manual text is **CC BY-SA 4.0**. The fonts are under their respective open licenses (SIL Open Font License and Apache 2.0 where applicable). Paged.js is MIT.

The visual language is inspired by xerox zines, 1990s computer magazines, and cut-and-paste hardware manuals. **No reference text, logos, or artwork from any existing publication is reproduced.** The halftone images are derived from the Jotnar project's own samples and showcase assets.

SOURCE

The manual source lives beside this PDF:

```
book.toml
content/
data/ops.tsv
press/
theme/book.css
assets/
scripts/
```

Build it with:

```
cargo run --manifest-path press/Cargo.toml --release -- --preview
```

The Jotnar language lives in the parent repository. The Python implementation is the oracle. The parity harness is the witness.

NO SILENT GUESSES. NO REPRODUCED LOGOS. REBUILD THE BOOK.

COLOPHON RULE

END OF ISSUE 01

END